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圍道

The Art of Surrounding

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Preface

What's Go? A game, of course. But one might ask: How is it done precisely? What can be done and what not? Is it possible to extend its concepts to other games?

圍道 — *Wéi Dào* — *The Art of Surrounding* tries to answer such questions by constructing a rigorous theory of the game of Go, followed by its generalization on and out of the board.

Part I: Origins and format. We begin the study of the game by its history. We finish this part with the rules of the game and setting the motivation for the foundation.

Part II: Foundation. Now that we know the game, we are ready to build it formally. The two-dimensional version of the game is constructed. The board, the pieces, the colors, and the relations between them are defined, followed by the concepts of group, captures, and SuperKo. The legality of a play is formalized via a theorem. The SuperKo detection is handled by Zobrist Hashing. The whole structure is named Go, and we begin to generalize it. Later, we discuss *life in the game of Go*, also known as *unconditional life*, by interpreting, in this theory, the results of David Benson; also, we develop the theory around *conditional life* (WIP). At the very end we discuss two papers by Jin Bai Kim; in the second one of those there is a theorem that uses Benson's results, for which we develop a counterexample.

Part III: Abstraction. We now study the board from outside. Taking into consideration some intuition about three-dimensional Go, a concept about the quantity of liberties of a board naturally appears; such concept leads to a problem about liberties and how to control their growth whenever the board gets big (in dimension). We also attack one-dimensional Go, and finally we try to look beyond the game of Go in order to find if the structure we have built for it can suit other games.

I would like to thank Gusti Cabaña for being my first reader and giving me direction when I needed it most. Thanks also go to Diego Barrios and Eros Girardi for reviewing this work and providing invaluable feedback. Lastly, to Naza, for putting up with me throughout all these years and the years to come.

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合抱之木，
生於毫末；

九層之臺，
起於累土；

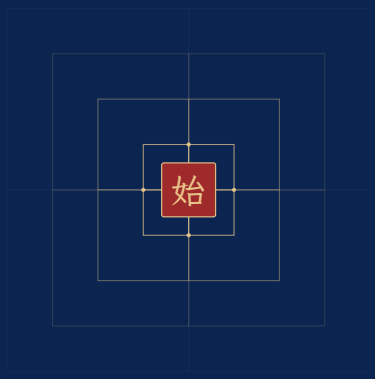
千里之行，
始於足下。

A tree as big as a man's embrace
grows from a tiny shoot;

A tower of nine storeys
begins with a heap of earth;

The journey of a thousand li
starts from where one stands.

道德經 — Dàodé jīng, Chapter 64



I: Origins and format

1.1 圍棋 (Wéi Qí): History and Rules

1.1.1 History of Go in China

The term 圍棋 (Wéi Qí) originates from the characters 圍 (Wéi), meaning ‘to surround’, and 棋 (Qí), which refers to a ‘board game’. In the West, it is commonly known as Go, a name derived from the Japanese 囲碁” (Igo). Although its exact chronology remains uncertain, ancient literary, pictorial, and archaeological records demonstrate that the game was an integral part of Chinese culture since BCE times.



Figure 1.1 – A game of Go on a traditional board.

Regarding its origins, mythical tradition holds that the legendary Emperor Yáo (堯) (2356 – 2257 BCE) invented it to enlighten the mind of his son, Dān Zhū (丹朱). Another account claims that Emperor Shùn (舜) (2255 – 2208 BCE) created it to educate his son, Shāng Jūn (商均). Furthermore, several sources attribute its invention to Wǔ, a vassal of Emperor Xià Jié (夏桀) (1818 – 1767 BCE), a courtier also celebrated for having devised the card game [15].

The first reliable mention of the game dates to the 3rd century BCE, when Mèngzǐ (孟子) describes Yì Qiū (弈秋), a master of the game during the Warring States period who was recognized as one of the foremost exponents of his time. On the other hand, a notable game is preserved in the Wàngyōu Qīnglè Jí (忘憂清樂集), a mid-12th-century compilation. This work records one of the earliest recorded games, dating from the Han period (2nd century BCE – 2nd century CE) and contested between General Sūn Cè (孙策) and his captain Lǚ Fàn (Figure 1.2) [6]. In conclusion, while it is unlikely that Go is 4,000 years old—as certain traditions suggest—its existence has been documented for at least 2,500 years. Its recurring appearance in texts and chronicles of the period suggests that, by then, the game enjoyed widespread enough familiarity for analogies based on it to be readily understood.

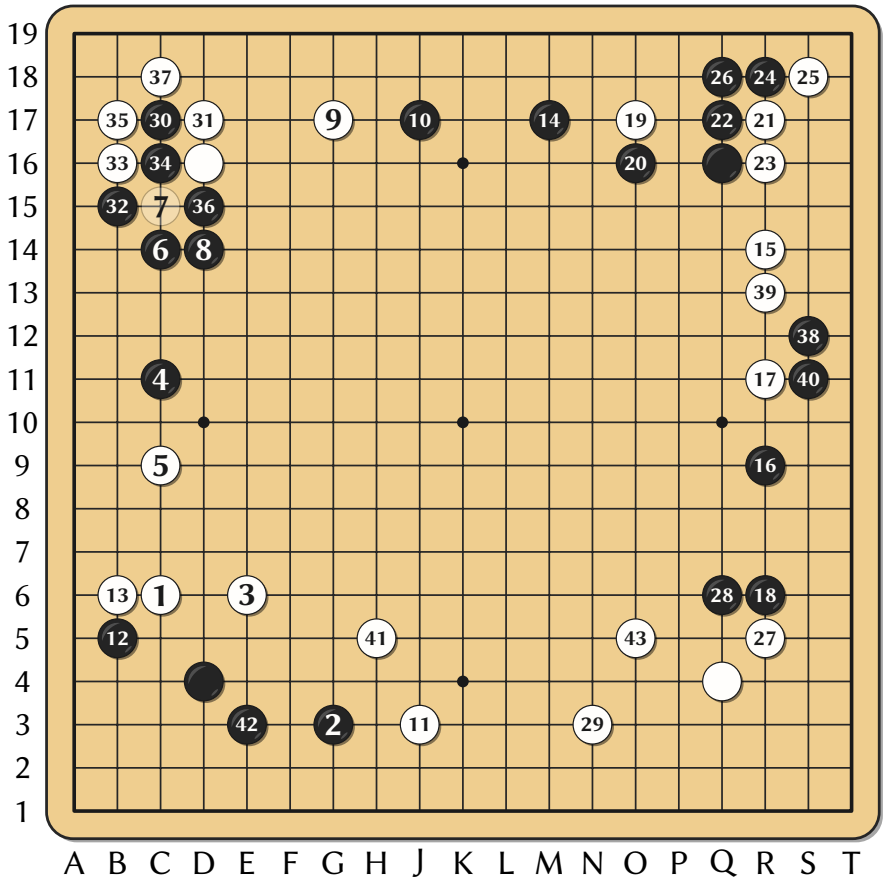


Figure 1.2 – Game between *Sūn Cè* and *Lǚ Fàn*. The unnumbered stones are due to the fact that, at that time, games began from a predetermined starting position and White played first.

1.1.2 History of Go in Argentina

Engineer Hilario Fernández Long was the pioneer of Go in Argentina. After learning the fundamentals from a Japanese friend, his enthusiasm led him to teach the first courses on the discipline at the Centro Argentino de Ingenieros (Argentine Center of Engineers). Thanks to his efforts, the Asociación Argentina de Go (Argentine Go Association) was founded on September 11, 1971. Later, in 1974, Fernández Long published the *Manual de Go* alongside Adalberto Moderc, which was followed in 1983 by the work *Introducción al Go* [13].

It is imperative to highlight the figure of Fernando Aguilar [1–3], Fernández Long's first cousin once removed and arguably the foremost exponent of Go in Argentina. Although he took his first steps in the world of strategy games through chess, under the tutelage of his father, Pedro Antonio Aguilar, Fernando discovered Go through the print media[1]:

> "At home, we used to buy the newspaper *La Nación*. Chess columns were published there regularly [...] At a certain point, articles about Go began to appear. The author was the same person who wrote the chess ones."

Driven by curiosity, he consulted his father, who replied:

> "Let's go visit your uncle Hilario so he can teach us how to play Go."

[1]The original quotes are in Spanish.

Aguilar, at the age of twelve [4], recalls his beginnings as follows:

> "Hilario explained the rules to us and gave us some general guidelines. As we parted, he gave us a board and stones set, as well as a book: *How to Play Go*, an English translation of a work by *Takagawa Shukaku* [...] From then on, the study sessions with my father consisted of reading a chapter of the book—which he translated for me from English—followed by a game played between us."

Having mastered the rules, it was time to compete.

> "The year was 1972, and Go activity across the country was in full swing. In September of that year, Master *Iwamoto Kaoru* visited our country."

Following the match between *Enrique Lindenbaum* and *Iwamoto*, it was the amateurs' turn:

> "Then came the simultaneous exhibition, where I had my first experience facing a professional player. I was at a beginner's level (or slightly above) and was supposed to take a nine-stone handicap, but *Sachiko Kondo* asked to play with seven stones [...] When it became evident that White's victory was overwhelming, the professional asked to suspend the game."[2]

Despite the defeat, *Fernando's* ambition only continued to grow:

> "The next step my father and I took was joining the Argentine Go Association [...] Soon after came the *Copa La Nación* tournament. Its second edition took place in 1972."

Although he did not have a standout performance that year, he certainly did the following year:

> "Well into 1973, the time came to contest the third edition of the *Copa La Nación* [...] The tournament was fiercely contested, but I managed to win it undefeated, relegating *Pedro Hecht* to second place."

The most significant milestone of his career came in 2002. After securing third place in the *Ibero-American Cup*, he qualified for the 1st *Toyota & Denso Cup – World Go Oza*. In the first round, on March 19, he faced *Hasegawa Sunao* (9*p*) (Figure 1.3). The opening proceeded customarily:

> "I grabbed a handful of stones, and my opponent guessed correctly that it was an odd number, so he was assigned to play with Black."[3]

Against all odds, the game ended in *Aguilar's* favor:

[2] Since Aguilar was the weaker player, he played first (taking Black), leaving *Kondo* to play with White.

[3] To determine who plays first in an even game, Japanese tradition employs a method called *Nigiri* (握り). One player (customarily the senior or higher-ranked player, though in international tournaments either player may hold the stones) takes a handful of white stones and conceals them in hand. The opponent then places one black stone on the board to guess an odd count, or two black stones to guess an even count. The white stones are then counted in pairs, if the guess is correct, the guessing player takes Black and moves first. Otherwise, the players swap colors.

> "The final margin was 3.5 points in my favor. The difference caught my attention, and it surprised Hasegawa: he asked for the game record and spent quite a while confirming that it was indeed so."

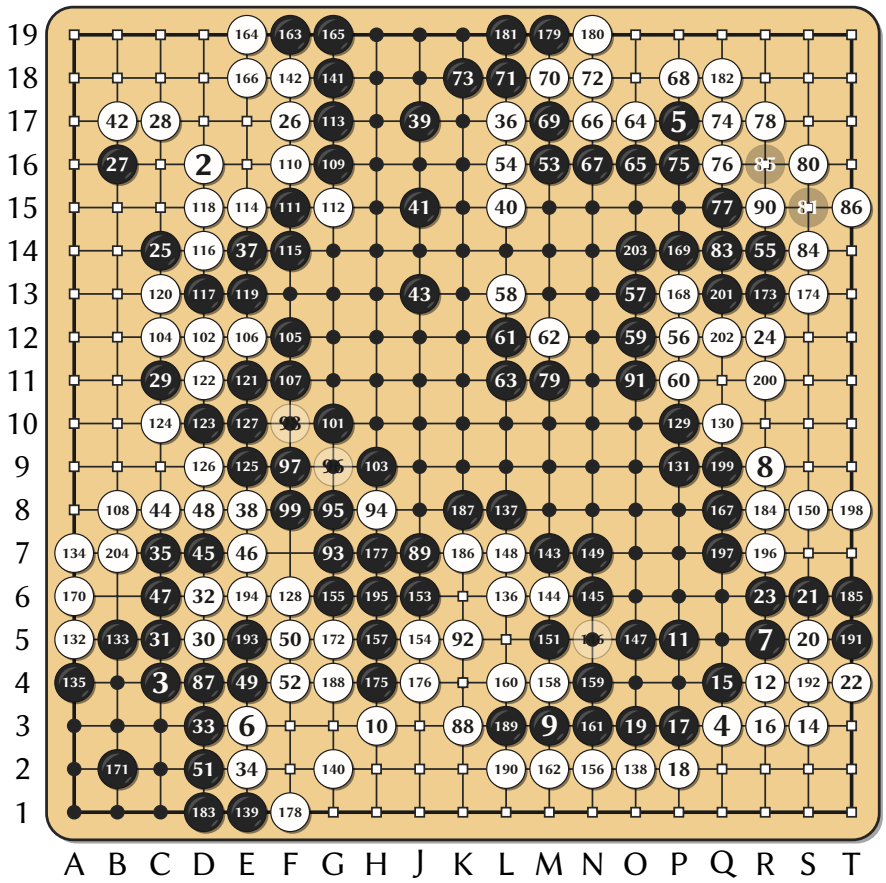


Figure 1.3 – Hasegawa Sunao (B) vs. Fernando Aguilar (W) – Toyota-Denso 2002, final board position. Each player *territory* is shaded. Captures are faded.

For the second round, held on September 2, Aguilar prepared exhaustively to face *Yo Kagen* (Figure 1.4). His preliminary analysis reveals the profound depth with which he perceived the game:

> "I was struck by the artistic manner in which he applies the 'surrounding principle.' His moves seemingly do not build anything specific [...] yet observing the board on a large scale, one notices that his groups form a grand enveloping maneuver [...] In a sense, I could say I 'fell in love' with *Yo's* style of play."

Before departing for *Tokyo, Japan*, where the tournament was hosted, *Fernando* recounts:

> "I left Formosa on Wednesday, August 28, and spent the night in Buenos Aires. There I saw my father, who saw me off saying: 'You already know that professionals are not invincible.' "

Once there, he managed to defeat *Yo Kagen*:

> "*Yo* appeared deeply disheartened upon realizing the outcome was decided. It took him about 40 minutes to compose himself, connect the final ko with White

268, and fill the dame in order to count [...] I felt that the most respectful thing to do was to wait in silence. We quickly sorted out the territories, confirmed the result, put away the stones, and *Yo Kagen* left quietly."

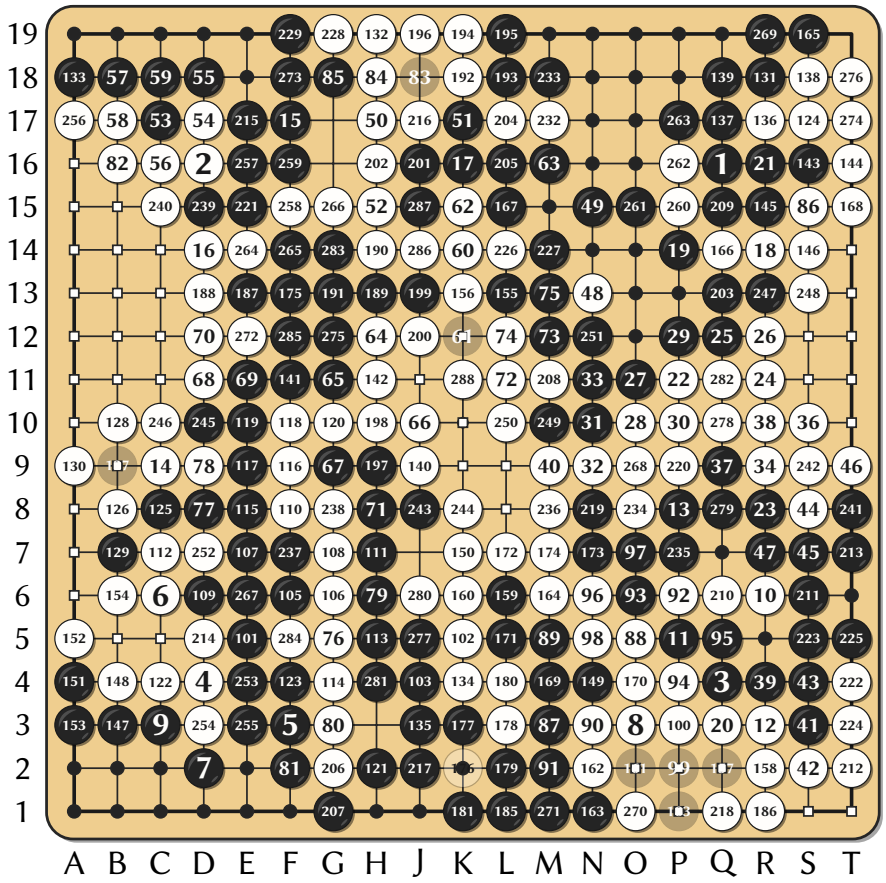


Figure 1.4 – Fernando Aguilar (B) vs. Yo Kagen (W) – Toyota-Denso 2002, final board position. Even if the reader cannot yet *read* the board, by the end of Part II they will be equipped with the necessary tools to do so.

Although he fell in the third round to the legendary *Lee Chang-ho* (Figure 1.5), Aguilar's feat remains a milestone of Western Go. Mention should also be made of his remarkable third-place finish at the 2008 *World Amateur Go Championship*. This brief excerpt scarcely does justice to Fernando's epic journey; readers interested in delving deeper into the game analyses are encouraged to consult [2, 3].

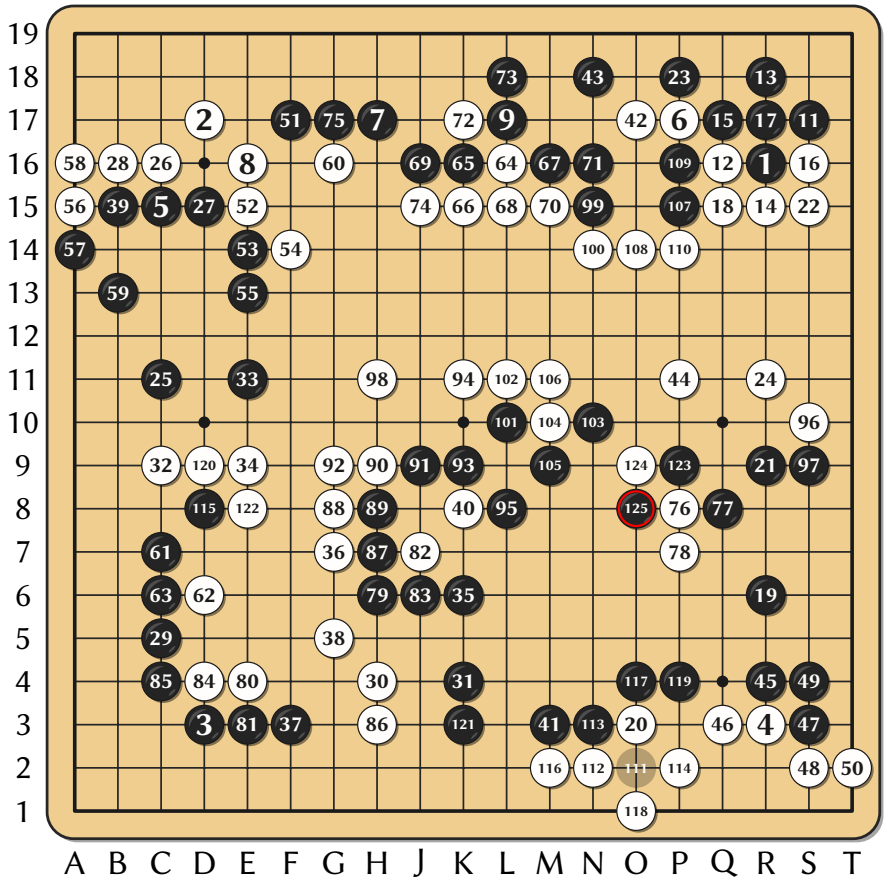


Figure 1.5 – Lee Chang-ho (B) vs. Fernando Aguilar (W) – Toyota-Denso 2002, final board position. Aguilar resigned after move 125.

1.1.3 History of Go Models

The vast literature on Go typically focuses on its complexity or on the search for optimal strategies, generally approaching them through *combinatorial game theory*. The present work belongs to the few studies devoted to *modeling* Go. This section is dedicated to tracing the historical development of such models.

The first paper exploring a potential mathematical model of Go was written by *Edward O. Thorp* and *William E. Walden* in 1964 [16]. In this article, the board is generalized from a fixed size to an $n \times m$ grid. Definitions of *adjacency*, the three states of any given intersection, and a basic formalization of the rules are established. Furthermore, they define Computer Go—a finite, cycle-free game—in order to study optimal strategies on small boards. They also introduce *t-truncated Go*, a variant that is forcibly terminated in a draw at turn t (if it has not ended earlier).

In 1970, *Albert L. Zobrist* published a method to encode a board position into a single large integer for state-verification purposes [18]. Although his work does not address Go per se, the method he describes will be employed in Section 2.4.1 to detect when one of the game's core rules, the *Superko* rule, is violated.

In 1972, Thorp and Walden published a combinatorial study exploring the nature and rules of Go to generate a computational model.

In 1976, David B. Benson [5] built upon Thorp and Walden's work to formally determine when a group is "alive". Although at first glance this contribution might seem separate from

general game models, it stands as a pioneering attempt to model the internal dynamics of Go. His approach will be revisited under our own theory in Section 2.6.

Finally, mention must be made of the work of *Jin Bai Kim*, conceived in 1976 and published in two parts in 1978 [8, 9]. This study continues the tradition of using graphs to model Go, although Kim omits one of the possible board states and proposes an alternative definition of *adjacencies*. In the second of these articles, we encounter the first formulation of three-dimensional Go (which we shall examine in Part III).

1.1.4 Rules and formats

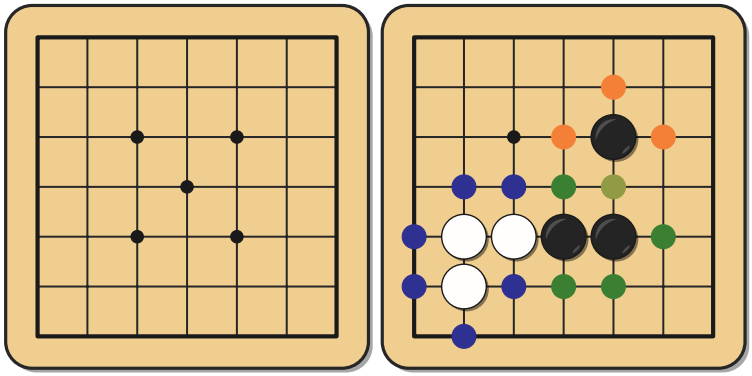


Figure 1.6 – An empty board, alongside a board illustrating the liberties of a single stone and of groups.

Go is played on a square board consisting of a grid of intersecting lines. It is contested by two players: one controls the black pieces and the other, the white ones. The most standard board sizes are 9×9, 13×13, and 19×19 (read as "nine-by-nine," "thirteen-by-thirteen," etc.). To introduce the rules, we shall employ a 7×7 board, a size more than adequate for this purpose.

As illustrated in Figure 1.2, coordinates are read starting from the lower-left corner, point $(1, 1)$, with coordinates increasing to the right and upward. The game begins with an empty board (Figure 1.6), unless played with a handicap. The small black dots visible at certain intersections serve merely as visual reference points and provide standard placements for handicap stones.

On their turn, a player may perform one of two actions: place a piece (termed a *stone*) on an unoccupied intersection, or pass. The player controlling the black stones plays first and may place their stone on any vacant intersection of their choosing.

Intersections on a board have associated *adjacencies*. These correspond to other intersections located at a distance of one unit in either the horizontal or vertical direction. Upon placing a stone on the board, it acquires *liberties*, defined as the unoccupied intersections that are *adjacent* to the stone or to a *group*. Two or more adjacent stones of the same color constitute a *group*; groups share the *liberties* of each individual stone comprising them (see Figure 1.6).

It can be observed that, on the second board, the black stone has four liberties, the white group possesses six liberties, and the black group has five. Throughout the game, a stone or group may be captured; this is accomplished by *completely surrounding* it, thereby removing

all the *liberties* it possesses. Figure 1.7 illustrates the capture of both a single stone and a group. When a capture occurs, the captured stones are removed from the board.

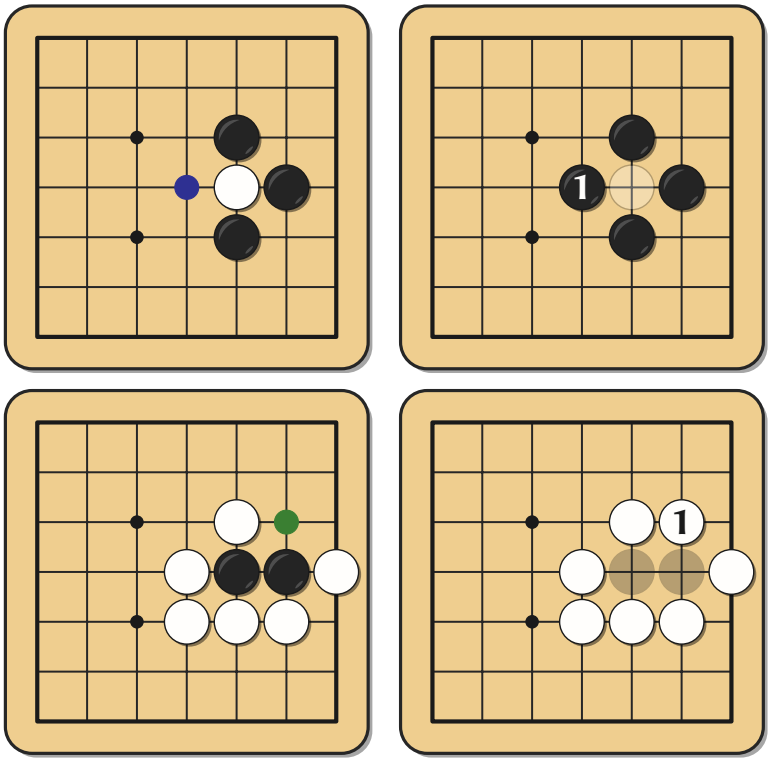


Figure 1.7 – Capture of a white stone and a black group.

For a stone to be placed on the board, it must have liberties at the intersection where it is played. The *suicide rule* prohibits placing stones on intersections that offer no liberties. Playing on such intersections would entail the immediate capture of the newly placed stone, tantamount to playing against oneself. Likewise, if a friendly group has only a single remaining liberty, filling that last vital space also constitutes suicide and is therefore an illegal move (see Figure 1.8).

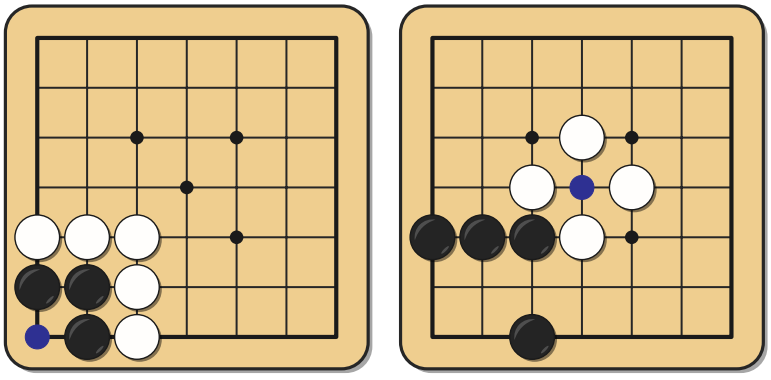


Figure 1.8 – In blue the illegal moves for Black as they are suicide.

There is, however, an exception to this rule: playing on an intersection with no liberties is legal if the move results in a capture. Upon removing the captured stones, the newly placed stone immediately gains liberties (see Figure 1.9). Furthermore, as seen in the figure, if both players were allowed to play on that intersection, the cycle of reciprocal captures could

continue indefinitely. To prevent this loop, the *ko rule* is introduced. If Black were to play at coordinates (4, 4) (blue dot in the figure), the board would revert to the exact position of the preceding turn.[4]

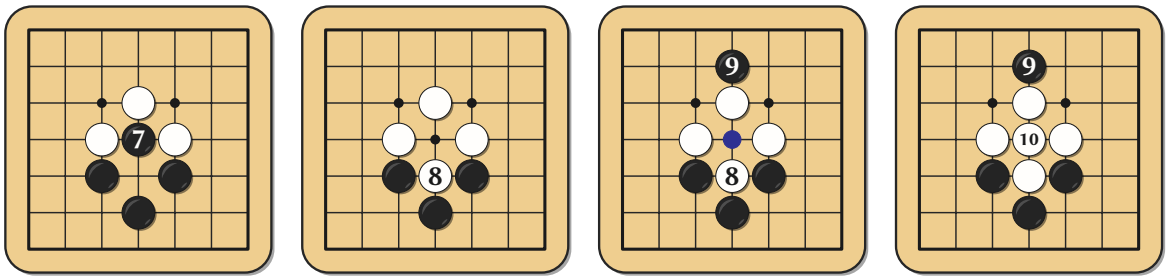


Figure 1.9 – An illegal position, as it reverts to a previous state. Following Black's move at (4, 6), Black could recapture at (4, 4) on a subsequent turn. White prevents this possibility by playing at (4, 4).

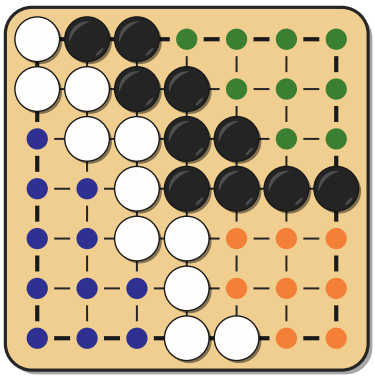


Figure 1.10 – Black territory, white territory and neutral.

To recapture, Black must first play elsewhere on the board (a ko threat). This generates a different board configuration; if the opponent responds, Black may then play at (4, 4) to recapture, as the board state will no longer be identical to the earlier one (see the final board in Figure 1.9). An empty intersection (or a group thereof) is considered the *territory* of a given color if it is completely surrounded by stones of that color. Conversely, if an empty space is bounded by stones of both players, it is regarded as neutral, belonging to neither player (see Figure 1.10).

When two players of different skill levels face each other, a handicap is usually granted to the less experienced player. This advantage consists of playing with the black stones (moving first) and placing a predetermined number of *handicap stones* on the board before the game begins; the greater the skill disparity, the more stones are placed (see Figure 1.11).

This is the traditional placement of handicap stones (on both 13 × 13 and 19 × 19 boards). There also exists *free handicap placement*, in which the weaker player freely chooses where to position these initial stones.

[4] We will see that the standard *ko rule* proves insufficient in certain scenarios, as longer cycles can occur that repeat a previous board position without violating this basic rule (see Section 2.2.1).

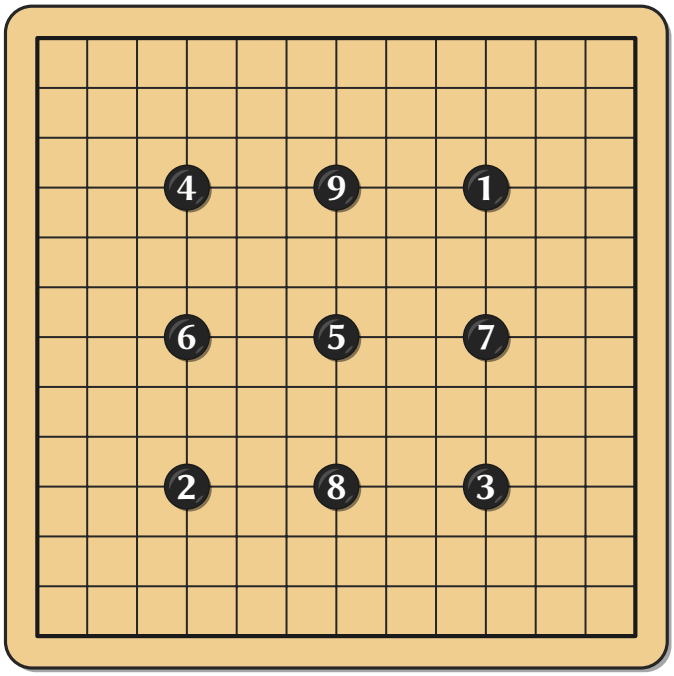


Figure 1.11 – Handicap of the 13 × 13 board.

The objective of Go is to secure more *territory* than the opponent. On their turn, a player may choose to pass, conceding the initiative to the rival. The game ends when both players pass consecutively. At that point, a phase of mutual agreement begins, during which all "dead" stones (that is, those whose eventual capture cannot be prevented) are identified and removed from the board.

After clearing these stones, the winner is determined by summing the secured *territories* and the living stones remaining on the board at the end of the game. Figure 1.12 illustrates a completed game.

The score would be computed as follows: 11 stones for Black plus 16 intersections in Black's territory; White, for their part, has 11 stones and 11 intersections in territory. Consequently, Black scores 27 points and White 22, so Black wins by 5 points.[5]

One might oppose Black's victory, given the inherent advantage of making the first move. To offset this disparity, White is awarded a compensation for playing second, known as *komi*.

This value varies according to board size and the skill difference between the players; generally, 5.5 points are used on standard boards.[6]

Were this compensation applied to the preceding example, White would win by 0.5 points, the narrowest possible margin. The inclusion of this half-point in the *komi* precludes the possibility of a draw.

A group possesses an *eye* if it completely surrounds an empty intersection. With at least two *eyes*, it achieves the state of being *alive* and can never be captured.

Consequently, the black group is *alive*. There exist, however, so-called false eyes: intersections that resemble *eyes*, but where one of the surrounding groups is at risk of being reduced

[5]This method corresponds to Chinese scoring rules. There are several other counting systems, such as the Japanese or Korean rulesets.

[6]The exact value of *komi* depends on the chosen ruleset. Furthermore, in handicap games, a minimal *komi* of 0.5 points is typically applied.

to a single liberty (the eye). As a result, if the opponent succeeds in capturing this vulnerable group, the structure collapses and the eye is lost (see Figure 1.13, right).

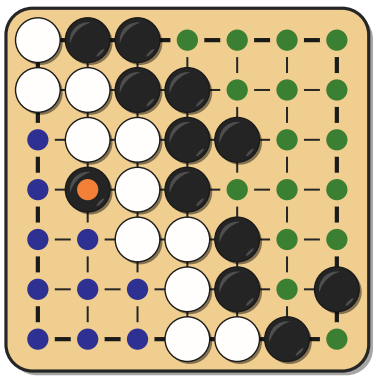


Figure 1.12 – Final board position of a completed game, with its intersections scored; the black stone marked with the orange dot must be removed as it is dead.

In Figure 1.13 (left), even if White attempts to attack the black group, they can only place one stone per turn. Because occupying either space would violate the suicide rule, White would need to play into both eyes simultaneously to capture, which is impossible.

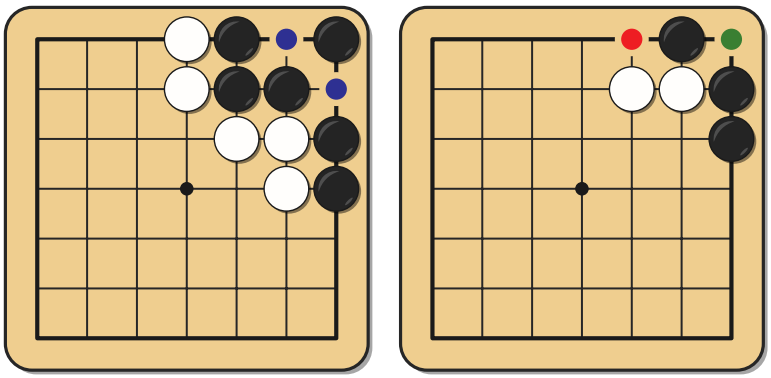


Figure 1.13 – Eyes of a living group (blue). False eye (green). If White plays at the red point, Black will be forced to fill their own eye with a stone, or risk losing it.

To build Go, we require four elements: *where*, *who*, *what*, and *how*.

三十輻共一轂，
當其無，有車之用。

埏埴以爲器，
當其無，有器之用。

鑿戶牖以爲室，
當其無，有室之用。

故有之以爲利，
無之以爲用。

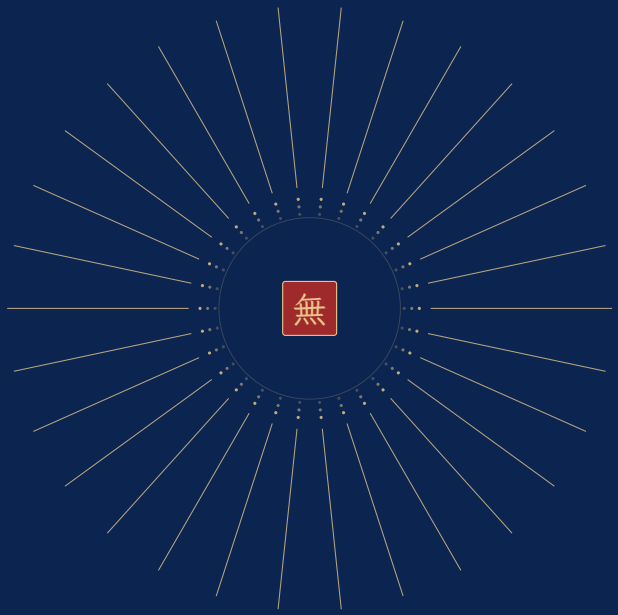
Thirty spokes converge to form a wheel hub.
But it is the emptiness inside the hub that makes it useful.

Clay is kneaded to form a pot.
But it is the emptiness inside it that makes it useful.

Doorways and window bays are chiselled to form a house.
But it is the emptiness inside it that makes it useful.

Substance brings benefits.
Emptiness brings usefulness.

道德經 — Dàodé Jīng, Chapter 11



II: Foundation

2.1 Go in two dimensions and two players.

Let us interpret the board in a convenient manner by considering just its points of intersection. Let $n, m \in \mathbb{N}$. We define $N = \{1, \dots, n\}$ and $M = \{1, \dots, m\}$. With these sets, we formalize the game space:[1]

$$G \coloneqq N \times M = \{(x, y) \,:\, x \in N,\ y \in M\}.$$

Let $t_1, t_2 \in G$ be two *points*[2]. If $t_1 = (a, b)$, we say that t_2 is adjacent to t_1, denoted by $t_1 \sim t_2$, if it satisfies exactly one of the following conditions:

$$t_2 = (a+1, b), \quad t_2 = (a, b+1),$$
$$t_2 = (a-1, b), \quad t_2 = (a, b-1).$$

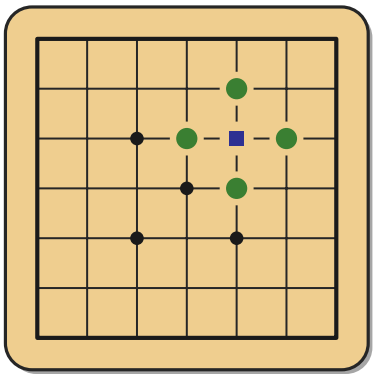


Figure 2.1 – Point $(5, 5)$ in blue, adjacencies in green.

If none of these conditions hold, t_1 is not adjacent to t_2, which we denote by $t_1 \nsim t_2$. Figure 2.1 illustrates the adjacencies of a point. Having established the *where*, let us examine *who* participates in it. When reviewing the rules, we observed that each point on the board can be in one of three possible states: unoccupied, occupied by a white stone, or occupied by a black stone. We define E as the set of states that a point may assume, $E = \{0, 1, 2\}$.

These values represent, respectively: vacant, black, and white. With this, we can account for all possible states on the board:

$$E \times G = \{(e, t) \,:\, e \in E, t \in G\}.$$

For example, the point $(4, 4) \in G$ may be found in one of three states: $(0, (4, 4))$, meaning no stone at $(4, 4)$; $(1, (4, 4))$, a black stone at $(4, 4)$; or $(2, (4, 4))$, a white stone at $(4, 4)$.

We shall refer to the elements of $E \times G$ as *stones*, denoted by the letter i, such that $i = (e, t)$,[3] with $e \in E$ and $t \in G$.

In Go, however, a single point cannot exhibit all states simultaneously; it assumes at most one.

Definition 2.1.1 (Board)**.** A subset $T \subseteq E \times G$ is said to be a *board* if,

$$\forall t \in G, \exists! e \in E \text{ such that } (e, t) \in T.$$

That is, in T there is a unique state for each position. Having established the *where* and *who*, we can construct the *what*. To do so, we require two fundamental functions:

Definition 2.1.2 (Stone State Function)**.**

$$\text{We define } C \,:\, E \times G \to E, \quad (e, t) \mapsto e.$$

[1] We use the letter G because we will adopt Japanese Go terminology for our notation whenever it does not lead to confusion. In Japanese, the board is called *Goban* (碁盤).

[2] The letter t originates from the term *ten* (点), meaning point in Japanese (and Chinese).

[3] From the term *ishi* (石), meaning "stone" in Japanese.

Definition 2.1.3 (Stone Position Function)**.**

$$\text{We define } P : E \times G \to G, \quad (e,t) \mapsto t.$$

Equipped with these, the notion of a group arises naturally:

Definition 2.1.4 (Connection)**.** Given a board T and $i_1, i_2 \in T$ two stones. We say that i_1 and i_2 are connected, denoted by $i_1 \leftrightarrow i_2$, if

$$C(i_1) = C(i_2) \quad \text{and} \quad P(i_1) \sim P(i_2).$$

Definition 2.1.5 (Group of Stones)**.** Let T be a board. Two stones $i_a, i_b \in T$ belong to the same connected component, denoted by $i_a \equiv i_b$, if $i_a = i_b$ or if there exists a finite tuple of stones $(i_1, i_2, \dots, i_k)$ in T $(k \geq 2)$ such that

$$i_1 = i_a, \quad i_k = i_b, \quad \text{and} \quad i_m \leftrightarrow i_{m+1} \quad \text{for all } 1 \leq m < k.$$

The relation $\equiv$ is an equivalence relation on T. Since connected stones share the same state, the state function C extends naturally to each equivalence class $I \in T/\equiv$ by setting $C(I) = e$, where $C(i) = e$ for all $i \in I$.

For each state $e \in E$, we define the family of components of state e on T as[4]

$$I^e := \{I \in T/\equiv \, : \, C(I) = e\}.$$

For $e \in E \setminus \{0\}$, the elements $I \in I^e$ are called *groups* of color e (when $|I| = 1$, $I = \{i\}$ corresponds to an *isolated stone*; when $|I| \geq 2$, I is a group). For $e = 0$, the elements $R \in I^0$ are called *regions*.

From now on, unless explicitly stated otherwise, the letter I will denote a group of non-zero color ($I \in I^e$ with $e \in E \setminus \{0\}$), whereas R will denote an empty region ($R \in I^0$).

Definition 2.1.6 (Adjacencies of a Group or Region)**.** Let $X \in T/\equiv$ be a group or region, and let $i = (e', t) \in T$ be a stone on the board. We say that X is adjacent to i, denoted by $X \sim i$, if at least one stone in X is adjacent to i:

$$X \sim i \iff \exists i_k \in X \, : \, P(i_k) \sim P(i).$$

It will prove very useful to identify the adjacencies of a stone, group, or region. To this end, we define the set K[5] associated with them (see Figure 2.2).

Definition 2.1.7 (Adjacency Set)**.** Given a board T, let $i \in T$ be a stone and let $X \in T/\equiv$ be a group or region. We define the adjacency set of the stone i as

$$K(i) = \{i_r \in T \, : \, P(i) \sim P(i_r)\}.$$

For a group or region $X \in T/\equiv$, we define

$$K(X) = \{i \in T \setminus X \, : \, X \sim i\}.$$

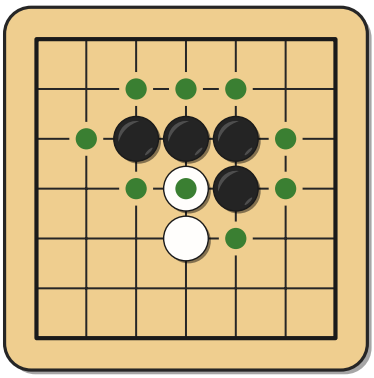


Figure 2.2 – A black group with its adjacencies marked.

[4] From *ishi* (石, stone/stones). Capitalized to signify sets of stones, and indexed by state e to denote the family of all such components on the board. When the board varies over time, we write $I^e(T)$.

[5] From the term *chikaku* (近く, near/nearby).

To characterize the states of these adjacencies, we apply the state function C to the adjacency set.

Definition 2.1.8 (State of the Adjacencies of a Group or Region)**.** Given a board T, let $X \in I^e$ be a group or region of state $e \in E$. Applying the function C elementwise to its adjacency set $K(X)$, we obtain the set of adjacent states, defined as

$$C(K(X)) := \{C(i) \mid i \in K(X)\} \subseteq E \setminus \{e\}.$$

This definition is essential for describing much of the internal dynamics of Go.

Definition 2.1.9 (Territory)**.** A region $R \in I^0$ is territory belonging to the player of color $e \in E \setminus \{0\}$ if $K(R) \neq \emptyset$ and there exists a non-empty family of groups $S \subseteq I^e$ such that

$$K(R) \subseteq \bigcup_{I \in S} I.$$

That is to say, all stones adjacent to the region belong to groups of the same color e. We shall denote the territories of each color as[6]

$$J^e := \{R \in I^0 \,:\, C(K(R)) = \{e\}\}, \quad e \neq 0.$$

Regions that do not constitute territory for any player will be denoted and defined as

$$J^0 := \{R \in I^0 \,:\, |C(K(R))| \neq 1\}.$$

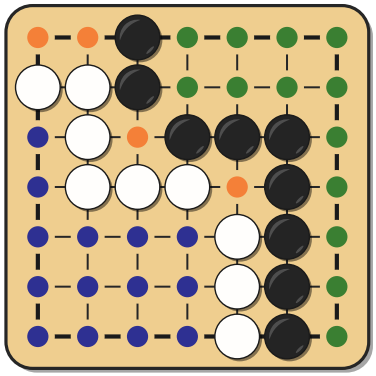


Figure 2.3 – White, black, and neutral territory.

Figure 2.3 illustrates the different types of territories. With this, on a two-player board we have established a partition of the set of regions I^0, since every region belongs to Black's territory, belongs to White's territory, or forms part of neither player's territory:

$$I^0 = J^0 \cup J^1 \cup J^2.$$

Additionally, the cardinality of the adjacency set of an individual stone, $|K(i)|$[7], proves remarkably useful for determining its spatial placement on the board. Because $|K(i)|$ is bounded by Lemma 2.3.1, a straightforward analysis of the number of adjacencies reveals precisely whether the stone is situated in the interior, on an edge, or in a corner. This geometric property will be essential, and we shall make use of it in Sections 3.2 and 3.4.

To formalize captures, we first introduce the concept of a liberty.

[6] From the term *ji* (地), meaning "land" in Japanese.

[7] This could be interpreted as the degree of a vertex in graph theory.

Definition 2.1.10 (Liberties of a Group)**.** Let $I \in I^e$ be a group of color $e \in E \setminus \{0\}$. We define the set of liberties of said group as

$$L(I) = \{i \in K(I) \,:\, C(i) = 0\}.$$

Note that if $L(I) = \emptyset$ for a given group, then that group is considered dead. This situation can only occur upon executing a move and effecting a capture. A group is captured when it loses all of its liberties, and captured stones are removed from the board.

Definition 2.1.11 (Capture Criterion)**.** Let $I \in I^e$ be a group of color $e \in E \setminus \{0\}$. Then,

$$I \text{ is captured} \iff L(I) = \emptyset.$$

We favor the phrasing *is captured*, since death is the consequence of a capture.

2.2 Toward Movement

Thus far, we have described the static structure of the board. To model a game, we must incorporate the temporal evolution of play. To this end, we first define the space of all possible moves, which we denote by $\mathcal{J}$:

$$\mathcal{J} := ((E \setminus \{0\}) \times G) \cup \{\emptyset\}.$$

2.2.1 Move Function and Legality

When playing Go on our turn, we either take a stone and place it on the board, or else we pass. We define the move function as $f : \mathbb{N} \to \mathcal{J}$. Let us examine an example. (See Figure 2.4)

$$\begin{aligned}
f(1) &= (1,(2,2)), & f(2) &= (2,(5,4)), & f(3) &= (1,(2,3)),\\
f(4) &= (2,(4,4)), & f(5) &= (1,(3,3)), & f(6) &= (2,(3,2)),\\
f(7) &= (1,(4,5)), & f(8) &= (2,(4,3)), & f(9) &= (1,(3,5)),\\
f(10) &= (2,(2,1)), & f(11) &= \emptyset, & f(12) &= \emptyset.
\end{aligned}$$

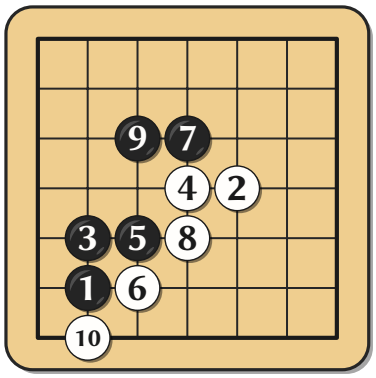


Figure 2.4 – Reconstructed game on a board.

The function records the moves, but not their legality. Let us analyze the rules. The *suicide rule* prohibits placing a stone if doing so results in the immediate capture of one's own group

(Figure 2.5). There exists a single exception: a move played with no liberties is legal if it simultaneously captures a group of the opposite color (see Figure 2.6). Note that, due to this mechanic, when it is Black's turn, only white groups may be captured, and vice versa.

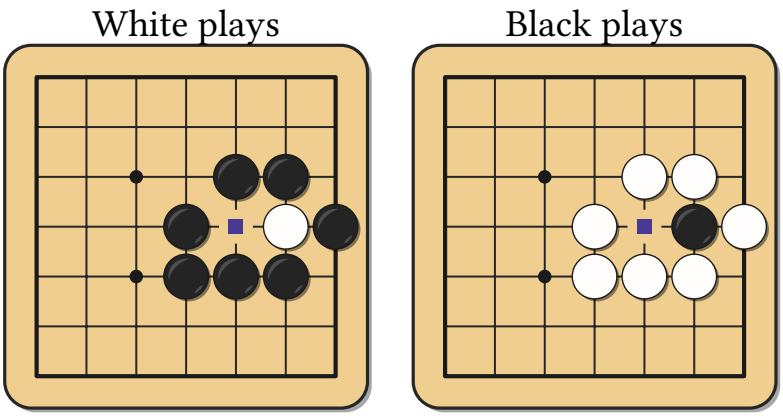


Figure 2.5 – In purple, illegal moves, as they lead to suicide.

To evaluate a move, we first verify whether it results in a capture. If so, the move is legal. If it does not capture, we inspect its liberties: if it has none, it constitutes suicide and is therefore illegal; otherwise, it is legal. As noted in Section 1.1.4, we must keep the *ko rule* in mind. This rule forbids the immediate recapture of a single stone if doing so reverts the board state precisely to the position of the preceding turn (see Figure 2.7).

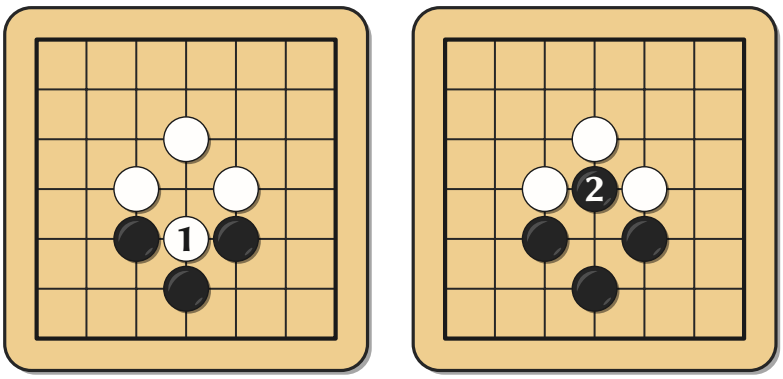


Figure 2.6 – Position of a stone under apparent suicide that is legal because it captures.

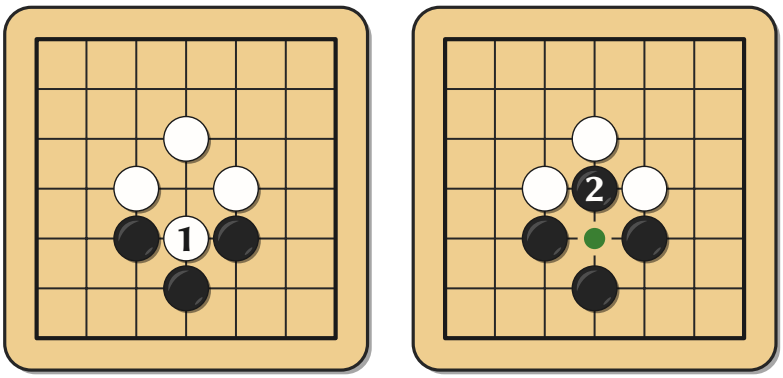


Figure 2.7 – White cannot play at the green intersection, as doing so repeats a prior board position.

The basic *ko rule* is insufficient, as cycles exist where the board position is repeated without immediately recapturing the same stone. See Figure 2.8, in which an earlier position is reached once again via a sequence of captures that do not violate the basic *ko rule*.

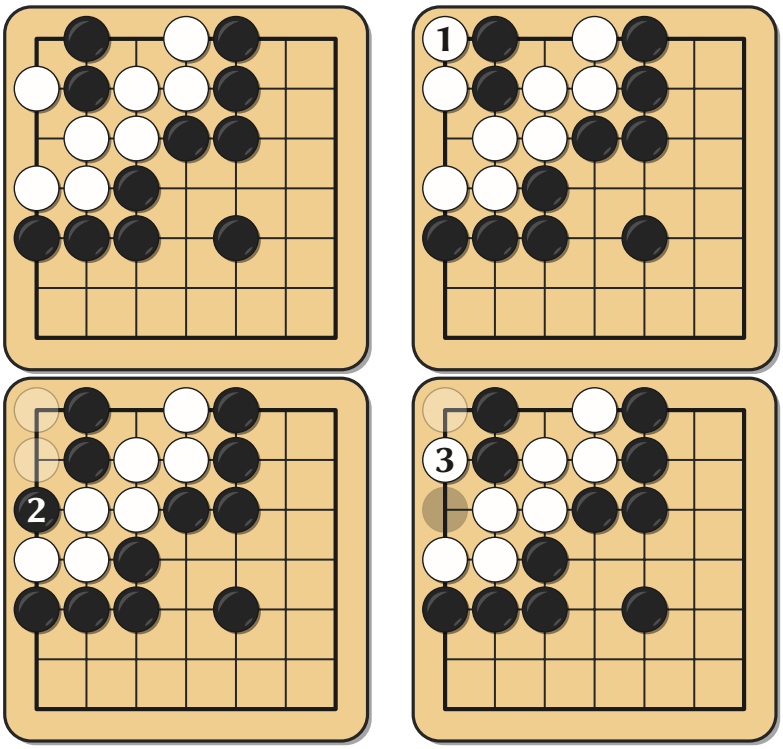


Figure 2.8 – Board configuration featuring a three-move cycle.

Because repeating positions is prohibited, we require a more general rule. To illustrate an extreme case, see Figure 2.9.[8]

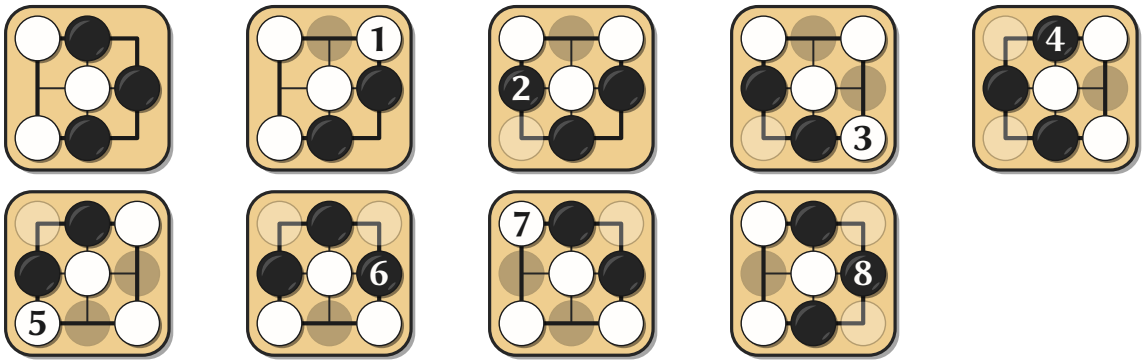


Figure 2.9 – A cycle of length 8 (assuming external group connections that leave the captured stones with their last liberty inside the 3×3 sub-region) that can be repeated indefinitely in the absence of a Superko rule.

One might argue that on the board in Figure 2.8, the position has indeed repeated, but the turn belongs to the other player; hence, it is not a full repetition. This line of reasoning leads to what is known as the *Natural Superko* rule. We, however, shall adopt a more restrictive variant:

[8] There exist examples involving far longer cycles; to explore them in greater depth, see [12].

Definition 2.2.1 (Positional Superko)**.** A player may not place a stone in such a way that it recreates a board position that has already appeared previously in the game.

By this rule, in both Figure 2.8 and Figure 2.9, any attempt to repeat a cycle will require intermediate moves that generate genuine game progress.

It remains to consider the most elementary restriction in Go: it is not permitted to play on an intersection that is already occupied by another stone.

Definition 2.2.2 (Vacant State Condition)**.** Let T_{n-1} be the board prior to turn n. For a candidate move $f(n) = (e, p)$ to be playable at turn n, the corresponding point p must previously be unoccupied in T_{n-1}. This is expressed as:

$$f(n) = (e, p) \text{ is playable on } T_{n-1} \iff (0, p) \in T_{n-1}.$$

Having established the *where*, *who*, and *what*, let us examine the *how*.

2.2.2 Legal Move Theorem

We have seen that a candidate stone $i = (e, p)$ is playable on T_{n-1} if $(0, p) \in T_{n-1}$. Let $T' = (T_{n-1} \setminus \{(0, p)\}) \cup \{i\}$ denote the tentative board obtained by placing i, and let $I(i) \in I^e(T')$ be the unique group of color e containing i on T'.

A move is non-suicidal if $L_{T'}(I(i)) \neq \emptyset$.

To detect a capture, we inspect T' to verify whether there exists any adjacent opposing group $X \in I^{e'}(T')$ (with $e' \in E \setminus \{0, e\}$) that is thereby left with no liberties.

This occurs if $L_{T'}(X) = \emptyset$; since the sole liberty remaining to that group was precisely the intersection just played, occupying it effects the capture. Had $L_{T_{n-1}}(X) = \emptyset$ held prior to the move, the group would already have been removed from the board.

Lastly, $f(n) = i$ produces a capture if and only if

$$\exists e' \in E \setminus \{0, e\},\ \exists X \in I^{e'}(T') \text{ with } X \cap K_{T'}(i) \neq \emptyset \ :\ L_{T'}(X) = \emptyset.$$

We define the Boolean functions Vacant, Suicide, Capture, and SKo. These functions take a candidate move $f(n) = i = (e, p)$ as their argument and return 1 if the condition is satisfied, and 0 otherwise:

$$\text{Vacant}(f(n)) = \begin{cases} 1 & \text{if } (0, p) \in T_{n-1}, \\ 0 & \text{if } (0, p) \notin T_{n-1}; \end{cases} \qquad \text{Suicide}(f(n)) = \begin{cases} 1 & \text{if } L_{T'}(I(i)) = \emptyset, \\ 0 & \text{if } L_{T'}(I(i)) \neq \emptyset; \end{cases}$$

$$\text{Capture}(f(n)) = \begin{cases} 1 & \text{if it captures on } T', \\ 0 & \text{otherwise;} \end{cases} \qquad \text{SKo}(f(n)) = \begin{cases} 1 & \text{if it breaks the rule,} \\ 0 & \text{otherwise.} \end{cases}$$

With these functions, we can formally characterize when a move is legal.

Theorem 2.2.1 (Legal Move Theorem)**.** *Let $f(n) = i \in (E \setminus \{0\}) \times G$ be a move at turn n of a Go game. Said move is legal if and only if it satisfies the following logical proposition:*

$$f(n) \textit{ is legal} \iff \textit{Vacant} \wedge \neg \textit{SKo} \wedge (\neg \textit{Suicide} \vee \textit{Capture}).$$

Proof. ($\Rightarrow$) Suppose that $f(n)$ is a legal move. By definition, a legal move does not violate the *Superko* rule; hence $\text{SKo}(f(n)) = 0$, and consequently $\neg\text{SKo}(f(n)) = 1$. Furthermore, we know that the *suicide rule* prohibits suicidal moves unless they result in a capture. That is, either the move is not a suicide, or (if it is) it necessarily effects a capture. In logical terms, since $f(n)$ is legal, the proposition $\neg\text{Suicide} \vee \text{Capture}$ is always true.

The move must be placed on a vacant point on the board; that is, $\text{Vacant}(f(n)) = 1$.

There exist only three categories of legal moves:

1. The intersection is vacant, there is no *Superko*, there is no suicide, and no stones are captured.
2. The intersection is vacant, there is no *Superko*, there is no suicide, and a capture occurs.
3. The intersection is vacant, there is no *Superko*, there is suicide, but a capture occurs.

($\Leftarrow$) Let us examine the truth table of our proposition. We denote Suicide by S and Capture by Cap for readability. We omit the cases where *Superko* is violated (SKo = 1) or where the intersection is occupied (Vacant = 0), as they are trivially false.

Vacant	$\neg$SKo	$\neg S$	Cap	$\neg S \vee$ Cap	Vacant $\wedge \neg$SKo $\wedge (\neg S \vee$ Cap)
1	1	1	1	1	1
1	1	1	0	1	1
1	1	0	1	1	1
1	1	0	0	0	0

The truth table demonstrates that the only rows rendering the full proposition true correspond precisely to the three configurations deduced in the forward direction. ●

Furthermore, we have characterized not only when a move is legal, but also which types of moves exist:

1. Placing a stone without capturing.
2. Executing a capture.
3. Executing a capture by means of suicide.

2.2.3 Scoring

Eventually, by virtue of Superko and mutual agreement, the game concludes. The winner in Go is the player who amasses the greatest number of points. To compute this, it suffices to sum each player's living stones remaining on the board at the end of the game and add their secured territory. In addition, the *komi* is added to White's total.

If we define $P(2)$ as White's score and $P(1)$ as Black's score, we obtain:

$$P(1) = \sum_{I_1 \in \mathscr{F}^1(T)} |I_1| + \sum_{J_1 \in J^1} |J_1|, \quad P(2) = \sum_{I_2 \in \mathscr{F}^2(T)} |I_2| + \sum_{J_2 \in J^2} |J_2| + komi.$$

To determine the winner, it suffices to evaluate the difference. If $P(2) - P(1) > 0$, White wins; otherwise, Black wins.

2.3 Go in n Dimensions and with m Players

If we transition to an n-dimensional space, we require n sets to construct the board. As before, for each dimension i, given $n_i \in \mathbb{N}$, we define $N_i = \{1, 2, \dots, n_i\}$. The board is then formed by the Cartesian product:

$$G = N_1 \times N_2 \times \cdots \times N_n.$$

For m players, we extend the set of states to $E = \{0, 1, \dots, m\}$, always reserving the value 0 for vacant intersections.

Definition 2.3.1 (Adjacency Between Points)**.** Let $G = N_1 \times N_2 \times \cdots \times N_n$ be the game space, and let $t_1, t_2 \in G$ be two points given by $t_1 = (a_1, a_2, \dots, a_n)$ and $t_2 = (b_1, b_2, \dots, b_n)$.

We say that t_1 is adjacent to t_2, denoted by $t_1 \sim t_2$, if:

$$\exists!\, i \in \{1, 2, \dots, n\} \,:\, \big(\forall r \neq i,\ a_r = b_r \ \wedge\ a_i = b_i \pm 1\big).$$

When reduced to two dimensions, the condition demands one identical coordinate and another differing by exactly 1, naturally recovering the definition of the classical board. Likewise, we extend the notion of connection between stones.

Definition 2.3.2 (Connection)**.** Let T be a board and let $i_1, i_2 \in T$ be two stones such that $i_1 = (e, t_1)$ and $i_2 = (e', t_2)$. These are connected if:

$$i_1 \leftrightarrow i_2 \iff t_1 \sim t_2 \text{ and } C(i_1) = C(i_2).$$

From now on, whenever we refer to this type of connection, we shall call it *orthogonal adjacency*, since we will see that it is possible to employ other criteria in Part III.

The definitions of groups remain identical, as they depend solely on the connection between stones. In turn, territories and captures function in exactly the same manner, since we only require the adjacencies of the group. The only element that truly changes upon introducing more players is the capture criterion; more specifically, the manner in which points are distributed upon removing opposing stones.

Before proceeding, let us highlight a geometric property of vital importance for the subsequent sections.

Lemma 2.3.1. *On an n-dimensional Go board $G = N_1 \times \cdots \times N_n$ with $n_i = |N_i| \geq 2$ for all $1 \leq i \leq n$ and orthogonal adjacency, a point $t = (a_1, a_2, \dots, a_n) \in G$ has at least n and at most $2n$ adjacencies.*

Proof. For each coordinate of t, there exists at least one valid direction, since if one is blocked by the edge, the opposite direction remains available (as $n_i \geq 2$). Consequently, each dimension contributes at least one adjacency. At an interior point, each coordinate contributes exactly two. Therefore, there are at least n adjacencies and at most $2n$. ∎

To simplify notation, we shall adopt the convention of denoting Go in n dimensions and with m players as Go_m^n. Should we wish to explicitly specify the size of each dimension, we may do so in the following manner: $\mathrm{Go}_2^2(19, 19)$ for the case of the standard board. If the number of players is omitted, it is to be understood that we are referring to a generic n-dimensional board, denoted by Go^n. Likewise, Go_m will represent a board of arbitrary[9] dimension played by m players.

[9] The infinite theory of Go is yet to be developed; I am referring to finite dimension.

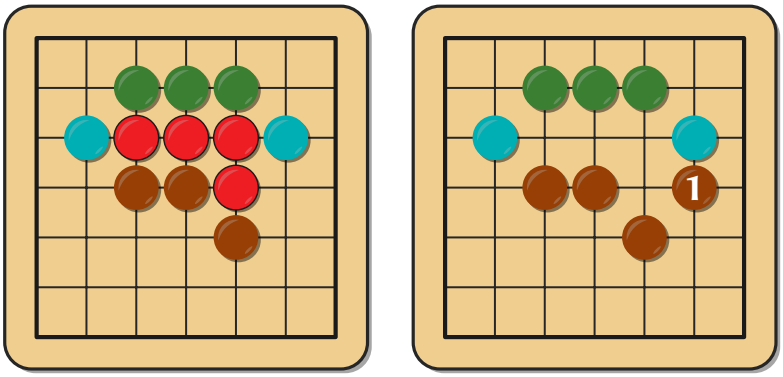


Figure 2.10 – Capture in Go_4^2. Brown plays and captures the red group.

2.3.1 Captures and Points for m Players

Let I be a group of stones. Just as on the classical board, for said group to be considered captured, it must hold that $L(I) = \emptyset$. Consider the Go_4^2 board illustrated in Figure 2.10. Let us suppose our players are: cyan (c), red (r), green (v), and brown (m). Our state set is then defined as $E = \{0, c, r, v, m\}$.

In this example, we observe that a group of 4 red stones is captured after the brown player places their stone at position $(6, 4)$. However, we note that the red group was not completely surrounded exclusively by brown stones. Of the 9 adjacent intersections (the liberties originally possessed by the red group), 3 have been occupied by the green player, 2 by the cyan player, and, finally, 4 by the brown player.

How do we distribute the capture points in this scenario? In general, when a group I is captured ($L(I) = \emptyset$), our proposal is to distribute its $|I|$ capture points among each attacking color $c \in E \setminus \{0, C(I)\}$ in strict proportion to the number of adjacent intersections $K_c(I) = \{i \in K(I) : C(i) = c\}$ occupied by that color:

$$\text{Points}_c(I) = \frac{|K_c(I)|}{|K(I)|} \cdot |I|.$$

If the total to be distributed is the 4 points corresponding to the red stones, we allocate them according to the fraction of liberties occupied by each attacker:

$$\text{Green's points} = \frac{3}{9} \cdot 4, \quad \text{Cyan's points} = \frac{2}{9} \cdot 4, \quad \text{Brown's points} = \frac{4}{9} \cdot 4.$$

Indeed, the sum of the parts equals the total number of points:

$$\left(\frac{3}{9} + \frac{2}{9} + \frac{4}{9}\right) \cdot 4 = 4.$$

In this manner, a proportional distribution is achieved, rewarding the player who contributes most to the capture. Awarding the points solely to the player who executes the capture would disincentivize aggression; it would be unfeasible to surround an opponent if, during the $m - 1$ intermediate turns, others could "steal" the capture.

2.3.2 Komi for Multiple Players

The origin of *komi* lies in preventing draws and counterbalancing the disadvantage of playing second; therefore, it is necessary to reinterpret this concept when multiple players are involved. What follows is a proposal on how we might generalize it.

Stepped Komi

We can define *komi* as a function that depends strictly on the turn order. If we consider the set of players $E \setminus \{0\}$, we can establish a compensation constant $\alpha \in \mathbb{Q}$. Thus, the *komi* function $K : E \setminus \{0\} \to \mathbb{R}$ is defined as:

$$K(j) = (j-1) \cdot \alpha.$$

Irrational Komi

If we additionally seek to preserve the second historical function of *komi*, preventing draws, we must ensure that no two players can achieve the same score. Let $Q_j \in \mathbb{Q}$ denote the rational score of player $j \in E \setminus \{0\}$ derived from stones, territories, and proportional captures, and let $\gamma_j \in \mathbb{R}$ be the *komi* assigned to player j. A tie between players j and k ($j \neq k$) occurs if and only if

$$Q_j + \gamma_j = Q_k + \gamma_k \iff \gamma_j - \gamma_k = Q_k - Q_j \in \mathbb{Q}.$$

Therefore, to guarantee the impossibility of ties, it is necessary and sufficient to assign *komi* values such that $\gamma_j - \gamma_k \notin \mathbb{Q}$ for all $j \neq k$. In particular, choosing an irrational constant $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ in the Stepped Komi formula $\gamma_j = (j-1)\alpha$ automatically satisfies $\gamma_j - \gamma_k = (j-k)\alpha \notin \mathbb{Q}$ for all $j \neq k$, eliminating any possibility of a tie.

Auction Komi

Prior to the start of the game, each player $j \in E \setminus \{0\}$ secretly declares a bid $b_j \in \mathbb{Q}$, representing the number of points they are willing to sacrifice from their final score in exchange for the privilege of securing the initiative (i.e., a negative *komi*). Once all bids have been collected, they are revealed. The turn order is established in strictly decreasing order: the player with the highest b_j value will play first, followed by the second highest, and so forth. In the event of a tie between two or more bids, the turn order for those players is resolved in favor of whoever placed their bid first.

These solutions may be employed simultaneously or further refined; what is presented here are ideas on how to rectify the lack of a well-defined *komi* in generalized Go.[10]

2.4 Positional Superko Detection

By virtue of the Legal Move Theorem 2.2.1, which by construction holds for n dimensions, we possess the theoretical tool to determine whether a move is legal; however, thus far we have not discussed the mechanism for its practical detection.

We have already formalized how to determine whether a capture or a suicide occurs, or whether the intended intersection is occupied. Now, when does a *Superko* occur? That is to say, how do we effectively ascertain that we have violated this rule?

A violation of the *Superko* rule implies that the candidate move recreates a board configuration that has already occurred in the game. To detect this, it would suffice to maintain

[10] Author's note: In my opinion, *komi* should not exist; the initiative (*sente*) can be lost at any moment. Furthermore, the larger the board, the less it matters who plays first. This 20th-century invention is only useful for professional players.

a historical record of the states of every intersection at each turn. Thus, by tentatively simulating the intended move and comparing the resulting state against the history, we could determine that the rule has been broken if an exact match exists.

To solve this problem, we shall make use of the work developed by Albert L. Zobrist [18].

2.4.1 Zobrist Hashing

For this method, we shall make use of the logical *exclusive or* operator, denoted as XOR ($\oplus$), applied to random numbers r_i represented as n-bit integers.

An n-bit integer is a sequence of n binary digits. For example, the number 15 in 4 bits is represented as 1111, since:

$$2^3 \cdot 1 + 2^2 \cdot 1 + 2^1 \cdot 1 + 2^0 \cdot 1 = 8 + 4 + 2 + 1 = 15.$$

Likewise, the number 9 in 4 bits is 1001. Let us recall the truth table for the logical XOR operator (see Figure 2.11).

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Figure 2.11 – XOR truth table.

Performing the XOR operation between 15 and 9 in base 2 means applying the operator individually to each corresponding bit. It is worth clarifying that, although we write the numbers contiguously, we actually treat them as if they were, in this case, 4-tuples: $1111 \oplus 1001 = 0110$. If r_i and r_j are independent, uniformly distributed random sequences of n-bit integers, the following properties hold:

P1) $(r_i \oplus r_j) \oplus r_k = r_i \oplus (r_j \oplus r_k)$. (Associativity)

P2) $r_i \oplus r_j = r_j \oplus r_i$. (Commutativity)

P3) $r_i \oplus r_i = \vec{0}$. (Nilpotence)

P4) If $s = r_1 \oplus r_2 \oplus \cdots \oplus r_k$, then s is also a random sequence.

P5) The resulting values s are uniformly distributed.

To begin constructing our algorithm, we must take the set of all possible configurations of a stone on the board, namely, $(E \setminus \{0\}) \times G$. We exclude 0 since we do not place stones of color 0 during play. Next, we assign a unique, random n-bit integer to each element. If we recall that a player has the option to *pass* their turn (a move that does not alter the board state), this action will be assigned $\vec{0}$, which guarantees that the state remains unchanged (as $\vec{0}$ is the neutral element of the sum).

To initialize the hash table so that all elementary stone-position pairs receive distinct non-zero seeds (plus $\vec{0}$ for the empty board), the minimum bit-length n must satisfy:

$$2^n \geq |(E \setminus \{0\}) \times G| + 1.$$

For our case, on the standard board, we will have 361 intersections and 2 possible states for the stones (white and black). Therefore:

$$|(E \setminus \{0\}) \times G| = 2 \times 361 = 722.$$

Hence, the minimum number of bits required to assign distinct seeds to the basic board table will be $n = 10$, since

$$2^{10} = 1024 \geq 722 + 1.$$

Zobrist warns of the existence of two types of errors. A *Type 1 error* occurs when two distinct board configurations produce exactly the same hash value; in the context of *Superko*, this would result in a false positive. On the other hand, a *Type 2 error* occurs due to memory space limitations, which we shall not consider. To reduce the probability of a Type 1 error, Zobrist proposes two complementary solutions. The first consists of generating a second hash simultaneously, utilizing an independent sequence, to rule out false positives. The second solution entails using a greater number of bits than strictly necessary. If $N = 2^n$ is the number of stored states we compare against and m is the number of extra bits added (so the hash length is $b = n + m$), Zobrist demonstrates that the probability of a new state avoiding a Type 1 error against 2^n states is given by the following expression (see Table 2.1):

$$p(\text{no error}) = \left(1 - \frac{1}{2^{n+m}}\right)^{2^n} \approx \left(\frac{1}{e}\right)^{2^{-m}},$$

m **(extra bits)**	**Prob. of no error**	m **(extra bits)**	**Prob. of no error**
0	0.36778	6	0.98449
1	0.60653	7	0.99222
2	0.77879	8	0.99611
3	0.88251	9	0.99805
4	0.93941	10	0.99901
5	0.96924		

Table 2.1 – Probability of no error by using more bits.

Therefore, it would suffice to utilize 21-bit or even 32-bit numbers to achieve a probability of avoiding a Type 1 error of 0.9999995 or 0.9999999997, respectively. We could generate two, three, or however many verification sequences we deem necessary in order to minimize the error.

Definition 2.4.1 (SuperKo detection)**.** Given a game trajectory $\{T_0, T_1, \ldots, T_{t-1}\}$, a play $f(t) \neq \emptyset$ breaks the SuperKo rule exactly when:

$$\text{SuperKo}(f(t)) = 1 \iff \exists k \in \{1, \ldots, t\} : T_t = T_{t-k}.$$

In the context of Zobrist, to verify the *Superko* rule, it suffices to perform at most $S \cdot t$ operations, where S is the total number of independent hash sequences implemented. If $H_s(t)$ represents the hash value at turn t for sequence s, then a stone placement $f(t) \neq \emptyset$ triggers a SuperKo detection when:

$$\text{SuperKo}(f(t)) = 1 \iff H_s(t) \oplus H_s(t-k) = \vec{0}$$

for some $k \in \{1, 2, \dots, t\}$ and for every sequence $s \in \{1, \dots, S\}$. We shall denote the numbers assigned to the stones by $h_s(i)$. Furthermore, to process captures, we shall make use of the XOR operator properties assumed previously. When a stone is captured, it is removed from the board. Therefore, if stone i_j is removed, it suffices to apply the operation $H_s(t) \oplus h_s(i_j)$ to the current state.

By virtue of Properties P1 and P2, we can rearrange the hash calculation:

$$H_s(t) \oplus h_s(i_j) = \big(h_s(i_1) \oplus \cdots \oplus h_s(i_j) \oplus \cdots \oplus h_s(i_m)\big) \oplus h_s(i_j)$$
$$H_s(t) \oplus h_s(i_j) = h_s(i_1) \oplus \cdots \oplus \big(h_s(i_j) \oplus h_s(i_j)\big) \oplus \cdots \oplus h_s(i_m) \qquad \text{(P1 and P2)}$$
$$H_s(t) \oplus h_s(i_j) = h_s(i_1) \oplus \cdots \oplus \vec{0} \oplus \cdots \oplus h_s(i_m). \qquad \text{(P3)}$$

Given that the null vector $\vec{0}$ is the identity element of the operation $(\vec{0} \oplus h(i_p) = h(i_p))$, the term vanishes. Thus, we have demonstrated that removing a stone (capturing) is equivalent to adding its hash once again. To conclude this chapter, let us illustrate with a concrete example on a small $\text{Go}_2^2(3, 3)$ board.

Since the board possesses 9 intersections, we have $|(E \setminus \{0\}) \times G| = 18$. Hence, the minimum number of bits n required will be 5, because $2^5 = 32 \geq 18 + 1$. If we decide to append 3 additional bits as a safety margin, we will have 8-bit numbers. To initialize, it suffices to randomly assign a number to each possible state at every intersection. To this end, we construct the following table:

Stone	Number	Stone	Number	Stone	Number
(1,(1,1))	01111111	(1,(2,1))	00010110	(1,(3,1))	01001101
(1,(1,2))	10001101	(1,(2,2))	11110001	(1,(3,2))	11101111
(1,(1,3))	10111110	(1,(2,3))	10110001	(1,(3,3))	00011110
(2,(1,1))	00001110	(2,(2,1))	00000010	(2,(3,1))	01001110
(2,(1,2))	10010011	(2,(2,2))	10111010	(2,(3,2))	01000111
(2,(1,3))	10110111	(2,(2,3))	10010001	(2,(3,3))	11001100

Table 2.2 – Hash table of each point in the 3 × 3 board.

We initialize the game hash to the null vector $\vec{0} = 00000000$, representing the empty board. From this point onward, for every executed move, we update the state by applying the XOR operation ($\oplus$) between the hash of the current turn and the value of the newly placed stone (or removed stone, in the event of a capture). Let us suppose that the following sequence of moves is executed, yielding the board shown in Figure 2.12:

$$f(1) = (1, (2, 2)), \quad f(2) = (2, (1, 3)), \quad f(3) = (1, (2, 3)), \quad f(4) = (2, (2, 1)),$$
$$f(5) = (1, (3, 1)), \quad f(6) = (2, (3, 2)), \quad f(7) = (1, (1, 2)), \quad f(8) = \emptyset,$$
$$f(9) = \emptyset.$$



Figure 2.12 – Finished game on a 3 × 3 board.

We shall compute the states turn by turn, partitioning the vectors into 4-bit blocks to facilitate readability. Denoting the hash of a stone by $h(p_i)$, where $p_i = (e, t)$, the evolution of the game proceeds as follows. On the sixth move, a capture is executed upon the black stone at $(3, 1)$; consequently, in addition to applying the hash of the newly played stone, we must apply the hash of the removed stone to "undo" its presence. Should multiple stones be captured, it would suffice to iterate the operation for each of them in any arbitrary order (by virtue of P1 and P2):

$$
\begin{aligned}
H(0) &= \vec{0} \\
H(1) &= H(0) \oplus h((1,(2,2))) = 1111\,0001 \\
H(2) &= H(1) \oplus h((2,(1,3))) = 0100\,0110 \\
H(3) &= H(2) \oplus h((1,(2,3))) = 1111\,0111 \\
H(4) &= H(3) \oplus h((2,(2,1))) = 1111\,0101 \\
H(5) &= H(4) \oplus h((1,(3,1))) = 1011\,1000 \\
H(6) &= H(5) \oplus h((2,(3,2))) \oplus h((1,(3,1))) = 1011\,0010 \\
H(7) &= H(6) \oplus h((1,(1,2))) \oplus h((2,(1,3))) = 1000\,1000 \\
H(8) &= H(7) \oplus 0000\,0000 = H(7) \\
H(9) &= H(8) \oplus 0000\,0000 = H(8).
\end{aligned}
$$

Figure 2.13 – Game hashes; moves in blue, captures in red.

At turns 8 and 9, passing the turn produces a state identical to the preceding one. This does not violate the *Superko* rule, as the rule solely prohibits placing a stone to recreate a position.

However, it could occur that $h((2,(3,2))) = 0000\,0010$. This would trigger a false positive for *Superko* on the sixth turn, since $H(6) = H(5) \oplus h((2,(3,2))) \oplus h((1,(3,1))) = 1111\,0111$ and, upon comparing it with the history,

$$H(6) \oplus H(3) = 0000\,0000.$$

Thus, by employing additional bits and implementing multiple simultaneous sequences to rule out collisions, Zobrist's method is fully implemented in Go.

2.5 The Structure of Time

We are now in a position to inject time into the structure. We define $\mathcal{T}$ as the space of all boards,

$$\mathcal{T} = \{T \subseteq E \times G \mid T \text{ is a board}\}.$$

2.5.1 The Game as a Trajectory

A game constitutes a sequence of moves $f : \mathbb{N} \to \mathcal{J}$, where $f(t)$ represents the action chosen by the active player at turn t. This function f naturally induces a sequence of boards in $\mathcal{T}$,

which we shall term a trajectory:

$$\{T_0, T_1, \dots, T_{\text{end}}\}$$

The sequence is constructed recursively by means of Δ:

1. **Initial state ($t = 0$):** T_0 is the empty board; that is, the set $T_0 = \{(0, t) : t \in G\}$.
2. **Transition ($t \to t + 1$):** The board at the subsequent turn is the result of applying the transition to the move of the current turn:

$$T_t = \Delta(T_{t-1}, f(t)).$$

2.5.2 State Dynamics

Under this construction, we formalize the properties of the game's flow. We shall denote by $E^k(T_t) = \{i \in T_t : C(i) = k\}$ the set of all stones of color k present on the board at turn t, and by $|E^k(T_t)|$ its cardinality.

- **Growth and Decrease:** If at turn t the active player is $k \in E \setminus \{0\}$, the new board given by $T_t = \Delta(T_{t-1}, f(t))$ will possess, at most, one additional stone of their color. In contrast, the stones of the other colors ($j \neq k$) can only decrease or remain constant due to potential captures:

$$\begin{aligned} |E^k(T_t)| &\leq |E^k(T_{t-1})| + 1, \\ |E^j(T_t)| &\leq |E^j(T_{t-1})|, \quad \forall j \neq k. \end{aligned}$$

- **Halting Condition:** Let $m = |E \setminus \{0\}|$ be the total number of players. The game concludes at turn $t_{\text{end}} \in \mathbb{N}$ when all players pass consecutively. This occurs when:

$$f(t_{\text{end}}) = f(t_{\text{end}} - 1) = \cdots = f(t_{\text{end}} - m + 1) = \emptyset.$$

  Consequently, the state of the board freezes, resulting in:

$$T_{\text{end}} \coloneqq T_{t_{\text{end}}} = T_{t_{\text{end}}-1} = \cdots = T_{t_{\text{end}}-m}.$$

  This state T_{end} constitutes the Final Board.

2.5.3 Anatomy of a Move

Let us understand how Δ functions. Suppose we are playing Go and that, on our turn, we intend to play $f(t)$. If $f(t) = \varnothing$, the board remains identical and we pass the turn; that is,

$$T_t = \Delta(T_{t-1}, \varnothing) = T_{t-1}.$$

If $f(t) = i^* = (e, p)$, we must validate said move. The computation of the new board $T_t = \Delta(T_{t-1}, i^*)$ by virtue of Theorem 2.2.1 is performed as follows:

1. **Placement:** If there exists $(e', p) \in T_{t-1}$ such that $e' \neq 0$, the move is illegal. If $e' = 0$, we generate a temporary board T' where intersection p, previously empty, acquires state e. We then proceed to the next step.

$$T' = (T_{t-1} \setminus \{(0, p)\}) \cup \{(e, p)\}.$$

2. **Capture Detection:** Evaluating the adjacencies in T', we identify the set Cap of all captured stones by taking the union of all opposing groups that possess no liberties:

$$\text{Cap} = \bigcup \left\{ X \in I^{e'}(T') \;\middle|\; e' \in E \setminus \{0, e\},\ L_{T'}(X) = \varnothing \right\}.$$

   If $\text{Cap} \neq \varnothing$, the move is exempt from the suicide check and we skip to step 4; otherwise, we proceed to the next step.

3. **Suicide Detection:** Considering $\text{Cap} = \varnothing$, let $I(i^*) \in I^e(T')$ be the group to which the played stone $i^* = (e, p)$ belongs on T' (whether $|I(i^*)| = 1$ as an isolated stone or $|I(i^*)| \geq 2$). If $L_{T'}(I(i^*)) = \varnothing$, the move constitutes a suicide and is therefore illegal. Otherwise, we proceed to the next step.

4. **Update:** We transform the captured stones to the free state (Cap_0) and update the board.

$$Cap_0 = \{(0, P(i)) : i \in Cap\}, \quad T_t = (T' \setminus Cap) \cup Cap_0.$$

   We then proceed to the next step.

5. **Superko Detection:** We compute $H(T_t)$ and compare it against the history. If the comparison yields $\vec{0}$ for any prior state (in every hash sequence), we have violated the rule and the move is discarded; otherwise, the move is legal and the turn concludes.

In this manner, any candidate move can be accepted or rejected. Denoting the adjacency relation on G by $A \subseteq G \times G$, a Game of Go is formally defined as a 4-tuple,

$$\Gamma = (G, E, A, \Delta),$$

where, to a Go structure (G, E, A), we adjoin the transition function $\Delta : \mathscr{T} \times \mathscr{J} \to \mathscr{T}$.

2.6 Life in the Game of Go

If we recall the rules, towards the very end we briefly encountered the concept of being *alive*. In this section, we shall formalize this notion.

In Figure 2.14, White cannot capture because the suicide rule prohibits them from playing on Black's remaining liberties. The algorithm for determining unconditional life is straightforward: we simulate Black passing while White attempts to occupy their liberties. If White fails to capture it, the group possesses unconditional life.

Now, formally,

Definition 2.6.1 (Unconditional Life)**.** Let $T_0 \in \mathfrak{T}$ be a board and let $S \subseteq I^e(T_0)$ be a non-empty set of groups of color $e \in E \setminus \{0\}$. We say that S possesses **unconditional life** on T_0 if there exists no finite sequence of legal moves $f(1), \ldots, f(n) \in \mathscr{F}$ $(n \in \mathbb{N})$, satisfying $f(t) = \varnothing$ whenever $t \equiv e \pmod{|E \setminus \{0\}|}$, such that some group $I \in S$ has no liberties on the tentative board T'_n $(L_{T'_n}(I) = \varnothing)$. In particular, a group $I \in I^e(T_0)$ possesses unconditional life on T_0 if $\{I\}$ possesses unconditional life on T_0.

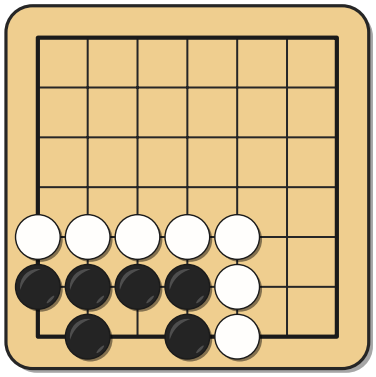


Figure 2.14 – Board with a black group that has unconditional life.

In 1976, David B. Benson formalized this concept [5]. He demonstrated that the survival of a set of stones depends solely on its adjacencies. In what follows, we shall adapt his definitions and theorems to our formal framework.

To achieve this, we first need to count how many groups surround a region. Let $P \subseteq I^e$ be a set of groups of color $e \in E \setminus \{0\}$, and let $R \in I^0$ be a region completely surrounded by P (that is, $K(R) \subseteq \bigcup_{I \in P} I$). We are interested in determining, knowing only R, how many distinct groups comprise its neighborhood.

We know that the set $K(R)$ contains the stones adjacent to the region. As seen in Section 2.1, $0 \notin C(K(R))$.

Definition 2.6.2 (Neighborhood of a region)**.** Let T be a board and let $R \in I^0$ be a region. We define the *Neighborhood* of R, denoted by ∂R, as the set of all groups (of non-zero color) that have a non-empty intersection with the adjacencies of the region:

$$\partial R := \left\{ I \in \bigcup_{e \in E \setminus \{0\}} I^e \;\middle|\; I \cap K(R) \neq \varnothing \right\}.$$

It is crucial to note the difference: ∂R collects the adjacent groups, whereas $K(R)$ collects the adjacent stones; Figure 2.15 illustrates this distinction. The cardinality $|\partial R|$ indicates exactly how many independent groups surround the empty region.

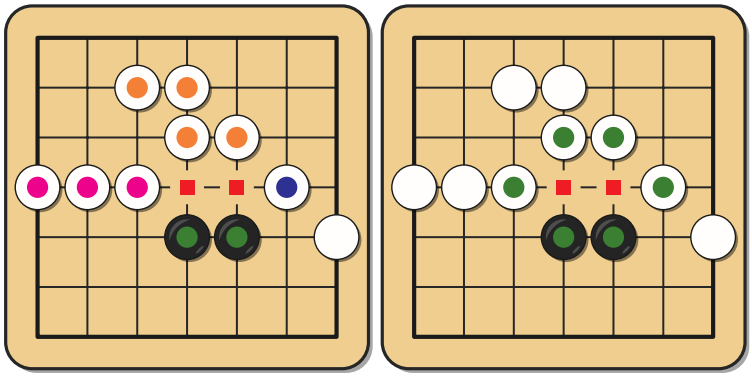


Figure 2.15 – Left: Board with the *neighborhood groups* of a Region; Orange, Magenta, Green, Blue. Right: Adjacencies of a Region.

However, quantity alone does not suffice; we must also evaluate the quality of these regions. In Benson's terminology, they must be *healthy* for the groups that border them.

Definition 2.6.3 (Healthy region)**.** Let $P \subseteq I^e$ be a set of groups of the same color $e \in E \setminus \{0\}$, and let $R \in I^0$ be a region surrounded by P (that is, $\partial R \subseteq P$). We say that R is *healthy* for a group $I \in P$ if every element of R is a liberty of I:

$$R \text{ is healthy for } I \in P \iff R \subseteq L(I).$$

This condition ensures that if the opponent attempts to play within the region, they will immediately come into contact with the group I. In the example of Figure 2.16, the intersections marked in green do not satisfy this premise; therefore, that region is not healthy for any black group.

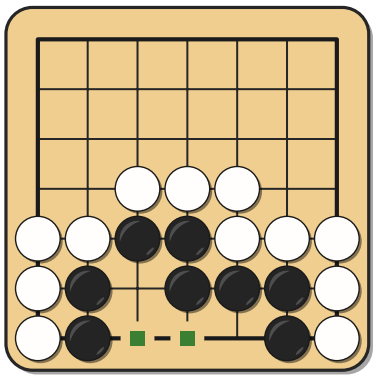


Figure 2.16 – Board with black groups whose controlled region is unhealthy.

Definition 2.6.4 (Vital region)**.** Let $P \subseteq I^e$ be a set of groups of the same color $e \in E \setminus \{0\}$, let $I \in P$ be a specific group, and let $R \in I^0$ be a region. We shall say that R is *vital* for I in P if the following conditions are simultaneously satisfied:

1. R is healthy for I: $R \subseteq L(I)$.

2. All groups comprising the neighborhood of R belong to P: $\partial R \subseteq P$.

We shall denote vital regions by V. This definition formalizes the requirement that a region must provide immediate liberties (be healthy) and be surrounded exclusively by groups in P. Building upon the notion of a vital region, we can classify the nature of completely surrounded regions.

Definition 2.6.5 (Eye)**.** Let $V \in I^0$ be a vital region for a group I within a set $P \subseteq I^e$. If the neighborhood of V is comprised of a single group—that is, if $|\partial V| = 1$—then V constitutes an *eye* for I.

When the region is surrounded by multiple distinct groups of the same color, it constitutes a joiner.

Definition 2.6.6 (Joiner)**.** Let $V \in I^0$ be a vital region for a group I within a set $P \subseteq I^e$ (so that $V \subseteq L(I)$ and $\partial V \subseteq P$). If the neighborhood of V contains more than one group in P—that is, if $|\partial V| > 1$—and V is vital for at least two distinct groups in ∂V, then V constitutes a *joiner* in P. Figure 2.17 illustrates the subtle difference between Benson's definitions.

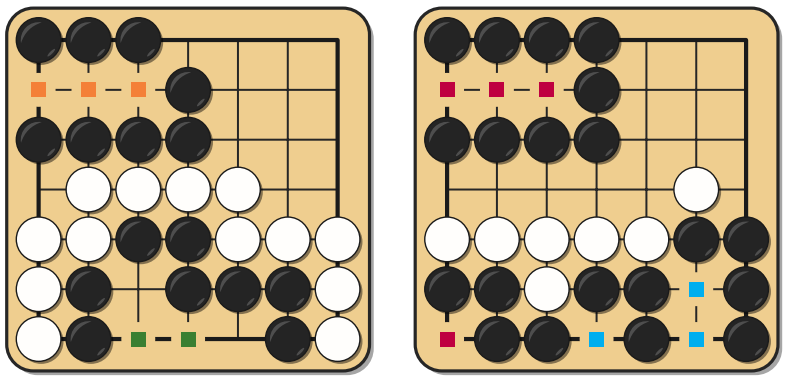


Figure 2.17 – Left: healthy and unhealthy. Right: eyes and joiners.

2.6.1 Benson's Theorem

Let us apply the theory and examine Benson's Theorem. Before proceeding to the main theorem, we must establish four preliminary lemmas. In what follows, we shall assume that the player defending the attacked group passes on every turn.

Lemma 2.6.1. *If a group I possesses at least two distinct eyes, then it possesses unconditional life.*

Proof. Let O_1 and O_2 be the eyes of I. To capture group I, the opponent must occupy all of its liberties, which entails completely filling O_1 and O_2.

Without loss of generality, let us suppose that the opponent attempts to fill O_1 first. Since the region is finite, there will eventually remain a single intersection $x \in O_1$.

Upon playing at x, the opponent's stones within O_1 are left with no liberties, as O_1 is exclusively surrounded by I ($|\partial O_1| = 1$). Given that I still possesses at least one liberty in O_2, the move at x does not capture I. Consequently, the suicide rule dictates that occupying intersection x is illegal.

As a result, it is impossible to completely eliminate either of the two eyes by means of legal moves. Therefore, I possesses unconditional life. ●

Lemma 2.6.2. *Let S be a set of groups of the same color such that $|S| \geq 2$. If a group $I_1 \in S$ possesses an eye O and at least one joiner V whose neighborhood contains another distinct group $I_2 \in S$, then I_2 must be captured before I_1 can be captured.*

Proof. To capture I_1, the opponent must occupy all the liberties provided by the joiner V, since those of the eye O can only be filled last (Lemma 2.6.1).

Given that V is a joiner, its neighborhood contains more than one group ($|\partial V| > 1$), including I_1 and I_2. Every move within V simultaneously reduces the liberties of both. Upon attempting to occupy the last empty liberty of V, the opponent's move will lack liberties of its own. By the suicide rule, this move is legal only if it executes a capture.

Since I_1 retains at least one liberty in its eye O, it cannot be captured on this turn. Therefore, the move must necessarily capture another group on the neighborhood of V, in this case I_2. Consequently, I_2 is captured before I_1. ●

In summary, to capture a group that possesses an eye and is connected to a joiner, it is necessary to first capture the other groups comprising the neighborhood of said joiner, as illustrated in Figure 2.18.

Lemma 2.6.3. *Let I be a group possessing no eyes, but connected to n joiners $V_1, V_2, \ldots, V_n$ ($n \geq 2$). Then, before I can be captured, the opponent must capture at least one distinct group in $n-1$ of these joiners.*

Proof. To capture I, the opponent must eliminate all of its liberties across the n joiners. Since they cannot simultaneously occupy the final liberty of every joiner, the suicide rule compels them to reserve at least one joiner, say, V_n, for the final move.

In order to exhaust the liberties in the $n-1$ remaining joiners ($V_1, \ldots, V_{n-1}$), each joiner presents the situation described in Lemma 2.6.2. Therefore, the opponent will be forced to capture at least one distinct group in each of those $n-1$ joiners before being able to attack I. ●

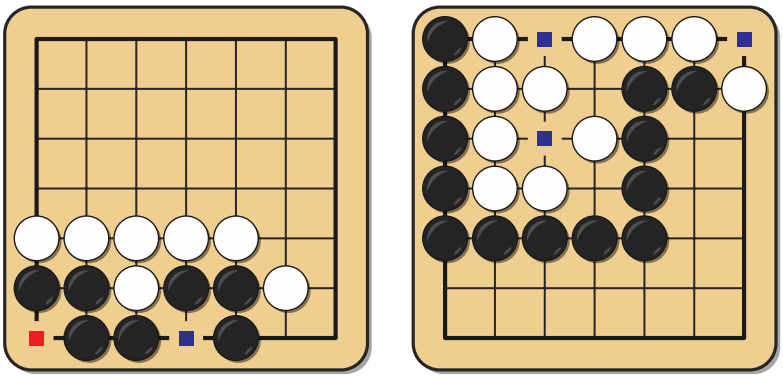


Figure 2.18 – Left: The black group possesses one eye (red) and one joiner (blue); White must first capture the group on the right (Lemma 2.6.2). Right: The upper-left white group lacks eyes, but shares joiners; Black must capture the neighboring groups before being able to capture it (Lemma 2.6.3).

Lemma 2.6.4. *Let $I \in P$ be a group or stone within a set P of groups of the same color; the liberties of I that do not belong to regions vital for I in P can be occupied by the opponent via a sequence of legal moves.*

Proof. Let $x \in L(I)$ be a liberty of I, and let R be the unique empty region such that $x \in R$. Suppose that R is not a vital region for I in P. By Definition 2.6.4, at least one of the two vitality conditions fails:

Case 1 ($\partial R \not\subseteq P$): There exists at least one group or stone in ∂R that does not belong to P (either an opposing group or a friendly group outside P that can be captured first). Since R is connected, the opponent can play stones across R starting from the boundary adjacent to $\partial R \setminus P$ (or leaving liberties within R), occupying the liberties $R \cap L(I)$ without suicide.

Case 2 ($\partial R \subseteq P$ but $R \not\subseteq L(I)$): Since R is not healthy for I, there exists at least one empty intersection $y \in R \setminus L(I)$ that is not a liberty of I (as in Figure 2.16). Because R is a connected region, the opponent can legally place stones along a path in R to occupy every intersection in $R \cap L(I)$ while leaving $y \in R \setminus L(I)$ vacant as a breathing liberty for the attacking chain, thereby avoiding suicide without needing to capture I first.

In both cases, all liberties of I outside its vital regions in P can be legally occupied. ●

Theorem 2.6.5 (Benson's Unconditional Life Theorem)**.** *Let I be a group. Under a canonical Go_m game ($m \geq 2$), I possesses unconditional life if and only if it belongs to a set of groups S of the same color such that, for every group $J \in S$, there exist at least two distinct vital regions in S.*

We shall begin by proving the backward implication. Then, via the contrapositive, the forward implication.

Proof. ($\Leftarrow$) : Let $I \in S$. To capture it, we must surround it completely. In particular, we must occupy the liberties of its vital regions. By Lemmas 2.6.1, 2.6.2, and 2.6.3, whether these regions are eyes or joiners, the capture of I demands the prior capture of some other group in S that belongs to the neighborhood of said regions.

Let us denote this group by $I_k \in S$. To capture it, we must eventually occupy the liberties of its own vital regions (which might coincide with those of I). To do so, once again, we must capture another group belonging to the neighborhood of its vital regions.

Repeating this argument, we deduce that every capture of a group in S requires a strictly prior capture of another group in S. Consequently, no finite sequence of moves can contain a first capture of a group in S. Therefore, S possesses unconditional life and, consequently, so does I.

($\Rightarrow$) : By contraposition, we shall prove that:

$$\exists I \in S \text{ with at most one vital region in } S \implies S \text{ does not possess unconditional life.}$$

Let $I \in S$ be a group possessing at most a single vital region V in S; by Lemma 2.6.4, it is possible to occupy all the external liberties of I by means of a sequence of legal moves. Once the external liberties are occupied, we analyze the state of I:

- If I lacked vital regions, the occupation of the exterior exhausts its liberties, thereby executing its capture.
- If I possessed exactly one vital region V, all of its liberties lie within V. Upon attempting to occupy the final intersection of V, since I does not possess a second vital region to guarantee it at least one liberty external to the move, placing this stone eliminates the final liberty of I, executing the capture.

In both cases, there exists a valid sequence to reduce the liberties of I and capture it. Therefore, S does not possess unconditional life; applying this elimination iteratively to any group with at most one vital region shows that any unconditionally alive group must belong to a non-empty surviving set S where every member has at least two vital regions, concluding the proof. ●

It is crucial to distinguish between being Alive and being Unconditionally Alive. In Figure 2.19, if Black were to play at the green intersection, they would achieve unconditional life; in its current state, the group lacks it. In a game, a group lacking unconditional life may be practically alive, requiring merely one or two moves to secure it. The distinction lies precisely therein: unconditional life implies that, statically, a group cannot die; life in Go, however, is a question of how quickly one can achieve it and whether the other players can prevent it.

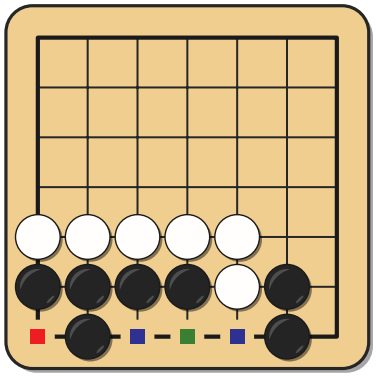


Figure 2.19 – Board with an eye and a joiner.

2.6.2 Seki

Having examined unconditional life, it is time to address conditional life. There exists another common structure in Go termed *seki*; this occurs when two groups of distinct colors share liberties in such a manner that, if either of them attempts to capture the other, the initiator ends up losing (see Figure 2.20). In Figure 2.20, whoever plays first and takes one of the two liberties of the interior groups loses, since it will no longer be their turn and the other player can (and most likely will) execute the capture.

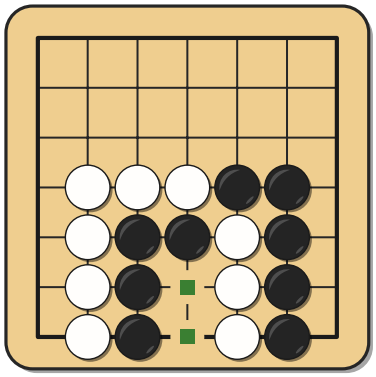


Figure 2.20 – Board with groups in *seki*.

Definition 2.6.7 (Basic Seki without eyes)**.** Let $I \in I^e$ and $J \in I^{e'}$ $(e \neq e')$ be two groups of distinct colors possessing no eyes. If there exists a shared region $R \subseteq L(I) \cap L(J)$ with $|R| = 2$ (or $|R| \geq 2$ such that reducing R reaches $|R| = 2$) satisfying $L(I) = L(J) = R$, then I and J are in seki.

This definition perfectly encapsulates *seki* without eyes, for if any player attempts to play in R, there will immediately remain a single liberty; whoever plays on the subsequent turn will execute the capture. It is necessary to require that $|R| = 2$; if $|R| = 1$, both groups would possess only a single liberty, and it would suffice for the active player to play on that liberty to execute a capture. By Benson's theorem, a group with two eyes possesses unconditional life.

It remains, then, to analyze the case of groups possessing a single eye that find themselves in *seki*.

Definition 2.6.8 (Basic Seki with one eye)**.** Let $I \in I^e$ and $J \in I^{e'}$ $(e \neq e')$ be two groups of distinct colors possessing single-point eyes O_I and O_J, respectively. If there exists a shared region $R \subseteq L(I) \cap L(J)$ with $|R| = 1$ (or $|R| \geq 1$ such that reducing R reaches $|R| = 1$) satisfying $L(I) = R \cup O_I$ and $L(J) = R \cup O_J$, then I and J are in *seki*. (See (Figure 2.21))

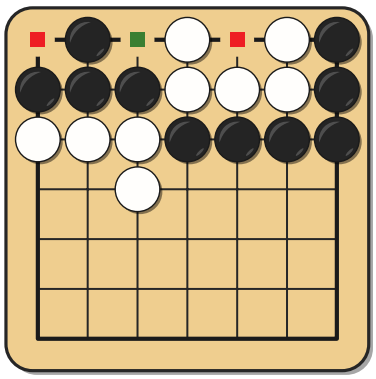


Figure 2.21 – Board with groups in *seki*.

By Lemma 2.6.1, every eye can be reduced to a single liberty. Consequently, occupying the shared region R entails relying exclusively on said eye, thereby allowing the opponent to capture the group. Although there exist more complex variants of *seki* (false eyes, double *ko*, etc.), these basic configurations suffice to illustrate conditional life.

This section will be continued some time in the future.

2.6.3 Remark on Kim's Theorem 2

We shall present a counterexample to Kim's Theorem 2 [9]. It is worth noting that Kim first raises the question of embedding an unconditionally alive group from a two-dimensional board into a three-dimensional board. However, Theorem 2 asserts a stronger result,

> "In a 2-person go-game on a 3-dimensional board, every two dimensional x-graph is not safe."

Given that Kim grounds his analysis in Benson's work, within our theoretical framework, this result translates as follows:

> "In a Go_2^3 game, no two-dimensional group of color x possesses unconditional life."

For Kim, a "two dimensional x-graph" is a group of color x that holds one of the spatial dimensions fixed at a constant value. It is worth noting that, in his formulation, he imposes a minimum board size of $9 \times 9 \times 9$.

We shall exhibit a configuration that contradicts Kim's Theorem 2.

Let $L_z = \{(x, y, z) : 1 \leq x, y \leq 9\}$ denote the z-th layer of the $9 \times 9 \times 9$ board. For every even z, let all positions in L_z be occupied by black stones, and denote the resulting group by B_z. For every odd z, leave L_z empty and denote the corresponding empty region by R_z. Thus, from bottom to top, the layers are

$$R_1, B_2, R_3, B_4, R_5, B_6, R_7, B_8, R_9.$$

Let $S = \{B_2, B_4, B_6, B_8\}$. Every R_z is a connected empty region. If z is odd, then every point of R_z is vertically adjacent to the black group immediately below it and, when $z \neq 1, 9$, also to the black group immediately above it. Consequently, R_z is healthy for each adjacent black group. Moreover, $\partial R_z \subseteq S$.

It follows that, for each $B_z \in S$, the regions R_{z-1} and R_{z+1} are two distinct vital regions. By Benson's Theorem, every $B_z \in S$ has unconditional life. Since every B_z fixes the third coordinate, it is a two-dimensional x-graph in Kim's sense. Therefore, the configuration contradicts Kim's Theorem 2.

Furthermore, if we omit the arbitrary board-size restriction imposed by Kim, the phenomenon can be illustrated in its minimal expression with the aid of the 3dGo tool.

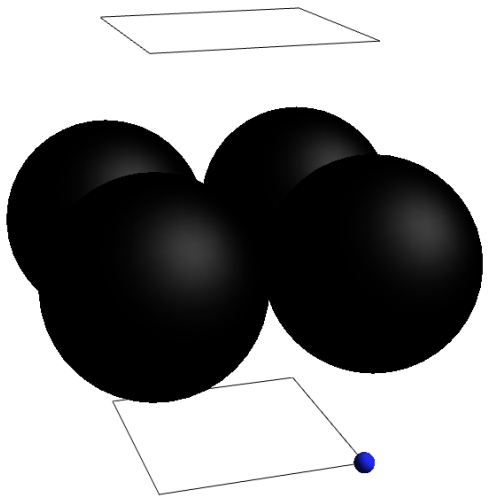


Figure 2.22 – Minimal counterexample to Kim's result. A two-dimensional black group possessing unconditional life on a $2 \times 2 \times 3$ board.

Behind the Curtain

Throughout this part, our goal has been to identify the minimal mathematical ingredients required to represent Go faithfully. We have shown that the game rests upon a four-component structure:

$$(G, E, \sim, \Delta),$$

where:

- G is the spatial abstraction of *where* we play.
- E is the abstraction of *who* plays, together with a neutral or vacant state (such as 0 in Go).
- $\sim$ is the relational abstraction of *how* positions are connected in space.
- Δ is the dynamical abstraction of a move, governing both its legality and the resulting state transition.

Thus, the triple $(G, E, \sim)$ defines the static arena of play, whereas the full tuple $(G, E, \sim, \Delta)$ constitutes the game itself.

In what follows, we shall examine the static structure $(G, E, \sim)$ from an external perspective before returning to the full dynamical system $(G, E, \sim, \Delta)$. Not only will this allow us to generalize the game of Go across dimensions and topologies, but it will also demonstrate that a much broader class of board games can be naturally understood within this same mathematical framework.

井蛙不可以語於海者，
拘於虛也；

夏蟲不可以語於冰者，
篤於時也；

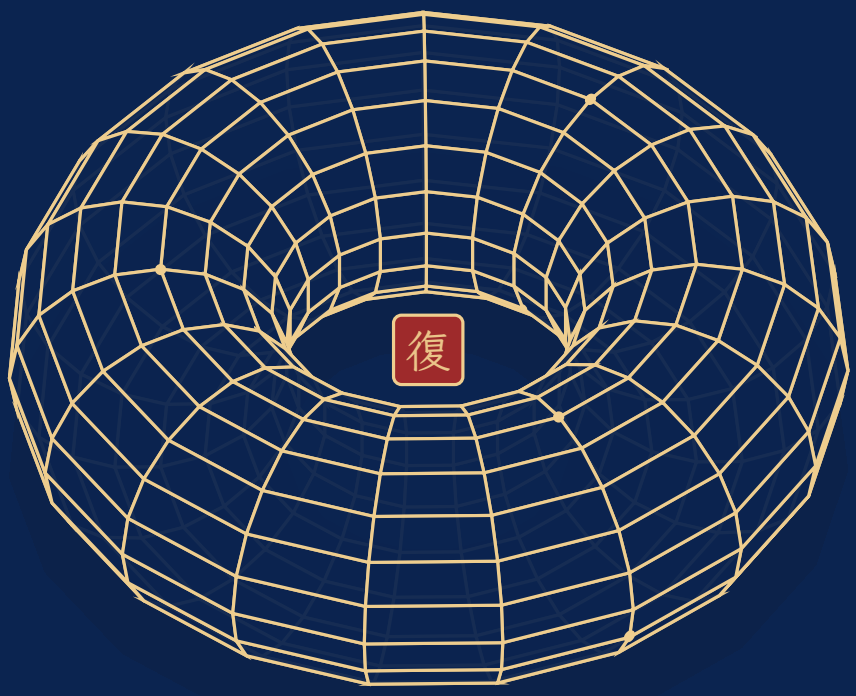
曲士不可以語於道者，
束於教也。

A frog in a well cannot be spoken to about the ocean
because it is confined by its own space.

A summer insect cannot be spoken to about ice
because it is bound by its own season.

A narrow-minded person cannot be spoken to about the Way
because they are bound by their own teachings.

秋水 — Zhuāngzǐ, Chapter 17



III: Abstraction

3.1 Comparing Adjacencies

Observing the two-dimensional board in Figure 3.1, we can note that it will have xy intersections; however, the following question arises: how many adjacencies do all these intersections possess?

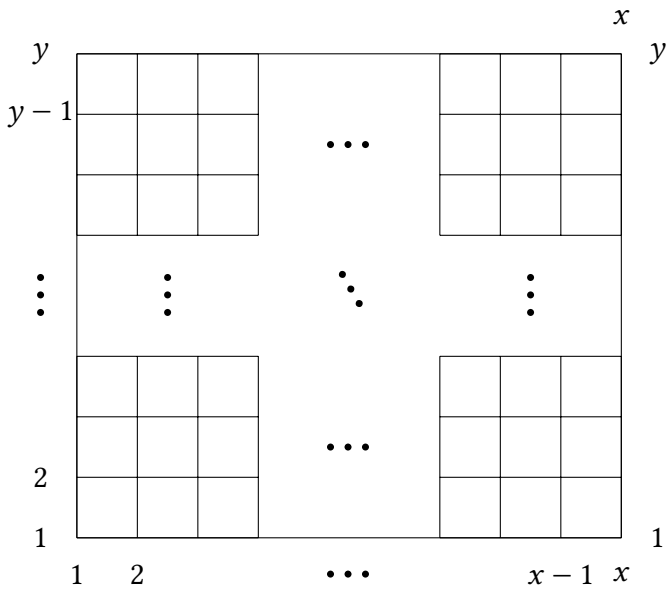


Figure 3.1 – Go board of size $x \times y$.

By Lemma 2.3.1, for $x, y \geq 2$, these nodes have 2, 3, or 4 adjacencies. The 4 vertices possess 2 adjacencies. There are four edges—two of length $(x-2)$ and two of length $(y-2)$—whose nodes have 3 adjacencies. Furthermore, there is an internal region of $(x-2)(y-2)$ nodes, each with 4 adjacencies. This allows us to compute the sum of vertex degrees (total directed adjacencies) on a Go^2 board.

Let $A(x, y) = \sum_{t \in G} |K(t)|$ be the total number of directed adjacencies on a board of dimensions $x \times y$ $(x, y \geq 2)$.

$$\begin{aligned} A(x,y) &= 2 \cdot 4 + 3(2(x-2)) + 3(2(y-2)) + 4(x-2)(y-2) \\ A(x,y) &= 8 + 6x - 12 + 6y - 12 + 4xy - 8x - 8y + 16 \\ A(x,y) &= 4xy - 2x - 2y. \end{aligned}$$

Let us also define the average number of adjacencies (average degree) on a board, which follows immediately from the definition of total adjacencies; it suffices to divide by the total number of nodes.

Let $\tilde{A}(x, y)$ be the average number of liberties of a single stone on a two-dimensional board, then:

$$\tilde{A}(x,y) = \frac{4xy - 2x - 2y}{xy}.$$

3.1.1 Go in Three Dimensions

Let us consider the case for Go^3, consisting of xyz nodes $(x, y, z \geq 2)$. To this end, we rely on the following diagram, Figure 3.2 (left):

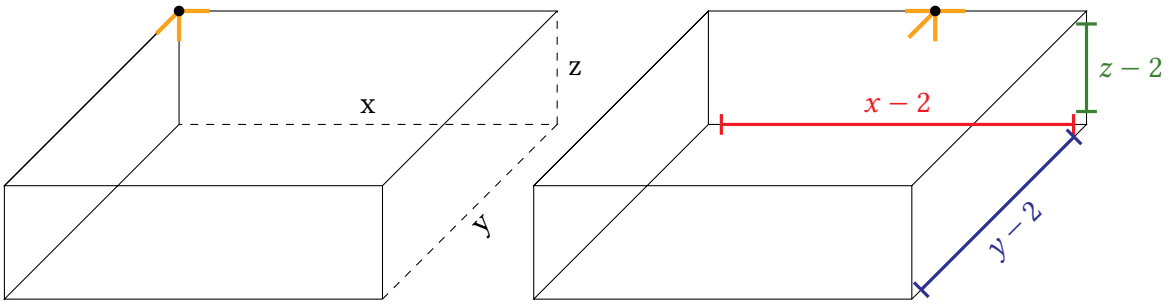


Figure 3.2 – Exterior of the three-dimensional board. A vertex with 3 adjacencies, edges, and a node with 4 adjacencies.

Proceeding as before, we observe that it possesses 8 vertices, each with 3 adjacencies. We identify 4 edges of $(x-2)$ nodes, 4 of $(y-2)$, and 4 of $(z-2)$, each with nodes possessing 4 adjacencies, see Figure 3.2 (right).

Next, we have the faces. These are divided into three pairs: 2 faces of $(x-2)(y-2)$, 2 of $(x-2)(z-2)$, and 2 of $(y-2)(z-2)$, with nodes possessing 5 adjacencies each, Figure 3.3.

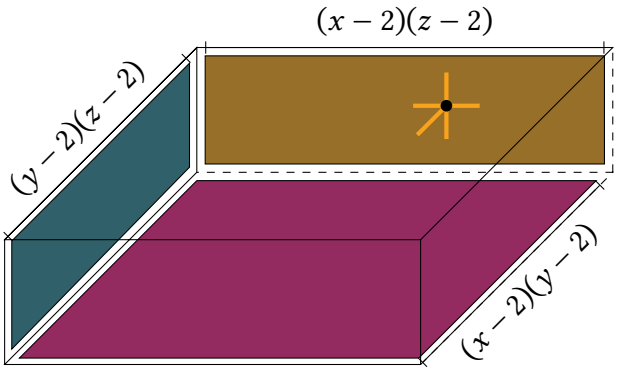


Figure 3.3 – Faces of the three-dimensional board with a node of 5 adjacencies.

Remaining is the internal volume, which consists of $(x-2)(y-2)(z-2)$ nodes, each with 6 adjacencies. We are thus in a position to define the total number of directed adjacencies on a three-dimensional board.

Let $A(x, y, z)$ be the number of directed adjacencies of a three-dimensional board; if:

- $V = 8 \cdot 3$, adjacencies of the vertices.
- $R = 4(4(x-2) + 4(y-2) + 4(z-2))$, adjacencies of the edges.
- $C = 5(2(x-2)(y-2) + 2(x-2)(z-2) + 2(y-2)(z-2))$, adjacencies of the faces.
- $I = 6(x-2)(y-2)(z-2)$, adjacencies in the interior.

In this manner, we can calculate the adjacencies of any three-dimensional board as follows:

$$A(x, y, z) = V + R + C + I$$
$$A(x, y, z) = 6xyz - 2xy - 2xz - 2yz.$$

And its average,

$$\tilde{A}(x, y, z) = \frac{6xyz - 2xy - 2xz - 2yz}{xyz}.$$

In turn, for $x, y, z \geq 3$, we can define the *exterior* of the three-dimensional board; let $A(x, y, z)_\partial$ be the number of adjacencies of a board of dimensions x, y, z that only takes the exterior into account. As seen previously, it suffices to sum the vertices, the edges, and, in the case of the faces, we no longer count the adjacency pointing toward the interior, since it ceases to exist as there are no interior nodes, such that our faces have 4 adjacencies. It suffices to take $C' = 4(2(x-2)(y-2) + 2(x-2)(z-2) + 2(y-2)(z-2))$, hence:

$$A(x, y, z)_\partial = V + R + C'$$
$$A(x, y, z)_\partial = 8(xy + xz - 2x + yz - 2y - 2z + 3).$$

We immediately define the average by subtracting the nodes we are not considering; let $\tilde{A}(x, y, z)_\partial$ be the average number of liberties on the exterior of a board of dimensions x, y, z:

$$\tilde{A}(x, y, z)_\partial = \frac{8(xy + xz - 2x + yz - 2y - 2z + 3)}{(xyz) - (x-2)(y-2)(z-2)}.$$

3.1.2 Comparing Average Adjacencies

Prior to the development of this work, as a Go player, I wondered whether a three-dimensional version of the game existed. Upon finding only conceptual images on the Internet, I asked myself: why not develop it? Thus, 3dGo was born.

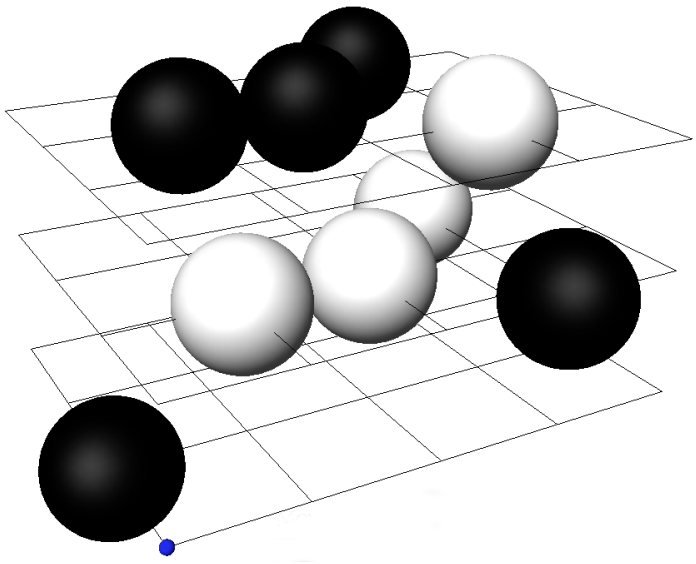


Figure 3.4 – Go board of size $5 \times 4 \times 3$ in 3dGo.

Once the rules, arbitrary board-size selection, and other options were implemented, my opponent and I noticed that it was not the same game. *3dGo* felt freer, more aggressive. The greater number of adjacencies altered the playing style, differentiating it from conventional Go. To understand this difference, even without prior experience, we shall analyze the behavior of the average number of adjacencies in the limit[1]: Let us examine the limits of the average adjacencies for two- and three-dimensional boards.

$$\lim_{x,y\to\infty} \tilde{A}(x,y) = \lim_{x,y\to\infty} \frac{4xy-2x-2y}{xy} = \lim_{x,y\to\infty} \left(4 - \frac{2x}{xy} - \frac{2y}{xy}\right) = 4,$$

$$\lim_{x,y,z\to\infty} \tilde{A}(x,y,z) = \lim_{x,y,z\to\infty} \frac{6xyz-2xy-2xz-2yz}{xyz} = 6.$$

It is evident that the average number of adjacencies in Go^3 is considerably higher. This difference is perceptible in the gameplay experience. The following problem then arises: How can we bring this average closer to 4? Playing Go^3 must be equivalent to playing Go; to achieve this, it must faithfully imitate it in rules, size, and gameplay feel. This is why this reduction is necessary, because playing Go^3 is not playing Go^2.

3.1.3 Removing Nodes, Yes. But How?

As we have seen, the majority of the nodes on a Go^3 board lie in the interior—provided the size is sufficiently large—where each possesses 6 adjacencies. Therefore, the elimination of nodes must begin in this region. Next in priority are the faces (5 adjacencies), whereas the edges and vertices, possessing 4 or fewer, do not affect the average we seek. If we consider only the exterior of the board, the faces possess precisely 4 adjacencies (we lose the one pointing inward); consequently, analyzing their limit, we find that:

$$\lim_{x,y,z\to\infty} \tilde{A}(x,y,z)_\partial = \lim_{x,y,z\to\infty} \frac{8(xy+xz-2x+yz-2y-2z+3)}{(xyz)-(x-2)(y-2)(z-2)} = 4.$$

Thus, our exterior board possesses the same average adjacencies in the limit as Go^2—that is:

$$\lim_{x,y\to\infty} \tilde{A}(x,y) = \lim_{x,y,z\to\infty} \tilde{A}(x,y,z)_\partial.$$

3.1.4 Toroidal Board

In Go^2, the limit of the average as the board is made arbitrarily large is 4. One might ask: is it possible to establish a board whose average liberties are precisely 4?

If we join the opposite edges of the board, we obtain the following, as shown in Figure 3.5.

[1] This was the motivation for initiating this writing.

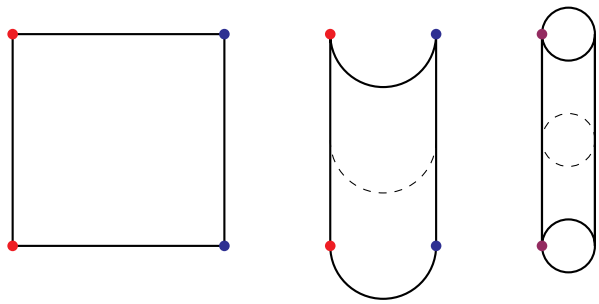


Figure 3.5 – Transition from the plane to a cylinder by joining its opposite sides.

Let us repeat the process; let us join our 4 vertices—which are now 2—into a single one, Figure 3.6.

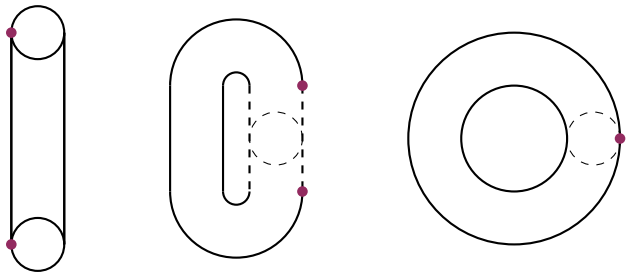


Figure 3.6 – Transition from the cylinder to the torus after joining its opposite faces.

Let us observe how the adjacencies of the vertices have changed: in the plane, we saw that they were 2 for each, but in the cylinder, it can be seen that they have 3, and furthermore, on the torus, they are 4. The same occurs with the nodes on the edges, Figure 3.7.

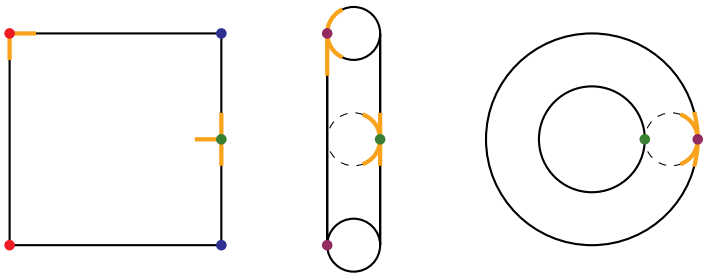


Figure 3.7 – Change in adjacencies from the plane to the torus.

It is therefore worth asking: Is it possible to do this in any dimension? That is, taking them to their "toroidal" form. Indeed, it is possible, and furthermore, it is very simple and, unlike Figure 3.7, we do not lose points.

Under our scrutiny, G contains only points, and the adjacency criterion determines their relationships. To relate the edges, it suffices to add new relationships. What we seek is to join each face with its opposite; to achieve this, we formulate the following definition of the adjacency criterion by opposites, denoted as A_{op}. Let $G = N_1 \times \cdots \times N_n$ be an n-dimensional board, where $|N_k| = n_k \geq 3$. We shall say that two points $g_1, g_2 \in G$ are oppositely related if

they are identical in all their coordinates except in one, the k-th, where they take values at the extremes:

$$g_1 = (x_1, \dots, 1, \dots, x_n) \sim (x_1, \dots, n_k, \dots, x_n) = g_2.$$

Thus, in this simple manner, we have defined the toroidal board, which is nothing more than the same board as before, except that we are now considering more adjacent points. Formally, if we call the canonical criterion A_{ort} (orthogonal adjacencies), we transition from the form (G, A_{ort}) to $(G, A_{ort} \cup A_{op})$.

It is clear that the toroidal board will have 4 adjacencies for each of its nodes. Let $A(x, y)_T$ be the number of adjacencies on a toroidal board of size xy; let us carry out once again the process we performed at the beginning. The toroidal board will have no vertices, no edges, and consequently, a single face composed of xy nodes which, as we have already seen, possess 4 adjacencies per node; thus:

$$A(x, y)_T = 4xy.$$

And its average will be 4, since:

$$\tilde{A}(x, y)_T = 4xy/xy = 4.$$

In this way, we have found a board such that for every node, the number of adjacencies is uniform. Let us do the same in Go^3; we can formulate the following scheme from Figure 3.8.

Here, if we apply the new relation, we obtain that all points will have 6 adjacencies. Indeed, the vertices will acquire 3 new adjacencies, bringing them to 6; the edges will acquire 2 new adjacencies, bringing them to 6; and finally, the faces will acquire a single new adjacency—the opposite one. Thus, equality is achieved in the limit.

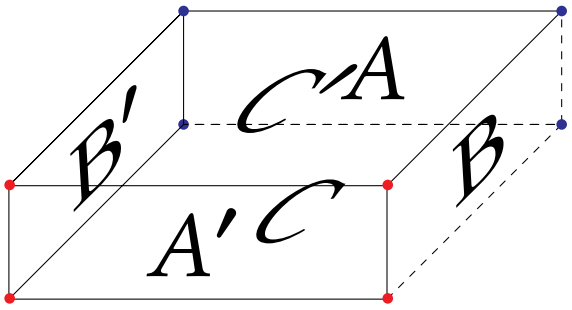


Figure 3.8 – Faces of the Go^3.

Now then, what would happen if we consider only the points lying on the boundary?

If $x \in G$, $x = (x_1, \dots, x_n)$ with $x_i \leq n_i = |N_i|$, the boundary will be that of the points given by

$$G' = \{x \in G : \exists i, (x_i = 1 \vee x_i = n_i)\}.$$

Through the canonical adjacency criterion, we obtain a board such that it possesses, in the limit, 4 adjacencies per point, since only the vertices possess 3 adjacencies, as we saw in Section 3.1.3.

If we attach to this board the adjacencies given by the opposite criterion, obtaining $(G', A_{ort} \cup A_{op})$, it will have 5 adjacencies for every face point (and 6 for edge and vertex points); since they lack the interior points, the faces lose one adjacency, but they gain it back

through the opposite face. The edges, which possess 4 adjacencies, gain 2, and the vertices obtain 3. Since the faces account for the largest number of points, it is clear that their limit will be 5, naturally, when scaled to large sizes.

If we recall Lemma 2.3.1, upon applying these procedures to any given board, we can always achieve the equality of $2n$ adjacencies per point, for the opposite criterion guarantees the filling of all missing adjacencies. It is easy to see that, by taking arbitrarily large boards, any n-dimensional board possesses an average of $2n$ adjacencies in the limit.

For this problem, for $n = 2$, A_{ort} provides a solution for the case with a degree as close to 4 as possible, whereas the relation $\mathscr{A} = A_{ort} \cup A_{op}$ satisfies the degree condition of exactly 4. Furthermore, for the case $n = 3$, we were able to find a partial solution on the boundary: if $G' = \{x \in G \mid \exists i \text{ such that } (x_i = 1 \vee x_i = n_i)\}$ denotes the boundary, then the graph $(G', A_{ort}|_{G'})$ is a solution for the case with a degree as close to 4 as possible on the surface.

3.2 Go in One Dimension

First, let us examine what it looks like, Figure 3.9. Clearly, there are two liberties per point, except at the boundaries, which possess one. Therefore,

$$A(n) = 2 \cdot 1 + (n-2) \cdot 2 = 2n - 2.$$
$$\tilde{A}(n) = \frac{2n-2}{n} = 2 - \frac{2}{n}.$$

Then, it is clear that the limit of its average is 2.

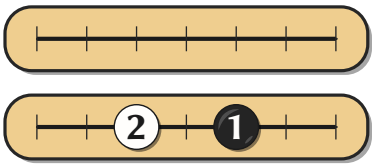


Figure 3.9 – Empty board and board with a black and a white stone.

Lemma 3.2.1. *In canonical* Go[1] *on* $G = \{1, \dots, n\}$ $(n \geq 3)$*, the only regions that can constitute an eye are the single-point extreme regions* $\{(0,1)\}$ *and* $\{(0,n)\}$*. Any other interior vital region is necessarily a single-point joiner.*

Proof. Every connected empty region R corresponds to a discrete interval of vacant points $P(R) = \{a, a+1, \dots, b\}$ with $1 \leq a \leq b \leq n$.

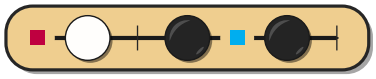


Figure 3.10 – Eye at the boundary, joiner in the interior.

- R **in the interior** $(1 < a \leq b < n)$**:** The adjacency set of R consists of the stones at $a-1$ and $b+1$. Since the interval $\{a, \dots, b\}$ is vacant, no path of stones of color $e \neq 0$ can connect $a-1$ to $b+1$. Thus, the stones $(e, a-1)$ and $(e, b+1)$ belong to two distinct groups $I_1 \neq I_2$, so $|\partial R| = 2$. Consequently, R cannot be an eye; if $b = a$ and $I_1, I_2 \in P$ share color e, $R = \{(0,a)\}$ is a joiner.

- R **at the boundary** $(a = 1$ **or** $b = n)$**:** Suppose $a = 1$ (the case $b = n$ is analogous). If $b = n$, the board is empty. If $b < n$, the only stone adjacent to R lies at $b+1$, belonging to a single group I ($|\partial R| = 1$). For R to be healthy for I ($R \subseteq L(I)$), every point in $P(R)$ must be adjacent to $b+1$, which forces $a = b = 1$. Hence $R = \{(0,1)\}$ is an eye for I.

●

3.3 Common Origin

Thus far, we have constructed two-dimensional Go and explored its generalizations to one and three dimensions. Although their dynamics differ, a natural question arises: are they truly distinct structures? We shall see that all of them can be derived from a single common origin.

A Go game is determined by a structure (G, E, A, Δ). To relate different variants, we shall introduce certain natural mappings that allow us to construct some variants from others while preserving the adjacencies and the rules of the game.

$$\phi : (G, E, A, \Delta) \to (G \times \{1\}, E, A_{G\times\{1\}}, \Delta_{G\times\{1\}}),$$

given by $\phi(g) = (g, 1)$. The preservation of adjacency is immediate; furthermore, ϕ is injective:

$$\phi(g_1) \sim \phi(g_2) \iff (g_1, 1) \sim (g_2, 1).$$

To expand the board with a finite set $M = \{1, \dots, k\}$, we define the dimension-expansion operator

$$\Xi_M : (G, E, A, \Delta) \mapsto (G \times M, E, A_{G\times M}, \Delta_{G\times M}).$$

Fixing a slice coordinate $m \in M$, the canonical embedding $\xi_m : G \hookrightarrow G \times M$ given by $\xi_m(g) = (g, m)$ preserves adjacency and is injective, since $(g_1, m) = (g_2, m) \implies g_1 = g_2$:

$$\xi_m(g_1) \sim \xi_m(g_2) \iff (g_1, m) \sim (g_2, m).$$

To incorporate additional players, we define an inclusion

$$\iota : (G, E, A, \Delta) \to (G, E \cup \{y\}, A, \Delta_{E\cup\{y\}}),$$

with $y \notin E$, such that $\iota(e) = e$ for all $e \in E$.

It is injective ($\iota(e_1) = \iota(e_2) \Rightarrow e_1 = e_2$) and preserves the identity of previous states. By design, it is non-surjective, so as to accommodate the new element.

Let us consider $\mathrm{Go}_0^0 = (\{1\}, \{0\}, A, \Delta)$, a single-node game with no players. Knowing that $\{1\} \times N \cong_A N$, we obtain the following diagram (Figure 3.11).

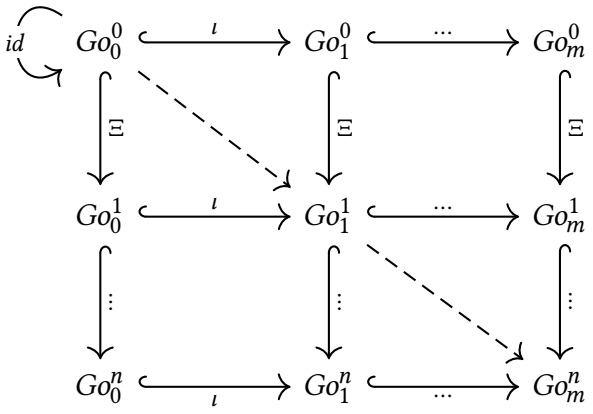


Figure 3.11 – Structure of the family of *Go Games*. Players are added to the right; the board is expanded downward.

Recalling 老子 — *Laozi*, we rediscover what has already been stated:

道生一， 一生二， 二生三， 三生万物。	*The way brings forth one,* *One brings forth two,* *Two brings forth three,* *Three brings forth all things.*

道德經 — Dàodé jīng Chapter 42

We can go a bit further; recalling what was seen in Section 3.1.4, we can define a natural transformation that associates each flat board with its toroidal version.

This mapping alters neither the number of nodes nor the players; it merely modifies the adjacency relations, identifying opposite positions on the board.

We thus define the toroidal mapping

$$\tau : (G, E, A, \Delta) \to (G, E, A \cup A_{op}, \Delta_\tau),$$

where A is the original adjacency relation and A_{op} is the relation between opposite positions.

The mapping acts as the identity on the nodes. Furthermore,

$$g_1 \sim_A g_2 \implies \tau(g_1) \sim_{A \cup A_{op}} \tau(g_2).$$

Since $A \subseteq A \cup A_{op}$, this property is immediate. In particular, τ is injective on the vertex set.

On the other hand, recalling the adjacency analysis carried out in Section 3.1, we observe that boundary positions exhibit a different behavior from that of interior positions.

To isolate this phenomenon, we introduce the boundary restriction operator

$$\partial : (G, E, A, \Delta) \to (\partial G, E, A|_{\partial G}, \Delta_{\partial G}),$$

where $\partial G = \{x \in G : \exists i, x_i = 1 \text{ or } x_i = n_i\}$ and $A|_{\partial G} = A \cap (\partial G \times \partial G)$, as we defined it in Section 3.1.4.

The Hollow Torus

Combining both constructions allows us to build the diagram of Figure 3.12.

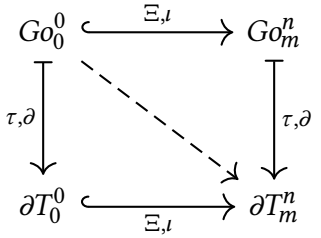


Figure 3.12 – Diagram of the complete structure. The horizontal arrows (Ξ) expand dimensions and players (ι). The vertical ones (τ) close the space, and (∂) hollow out the interior.

The Projection onto the Graph

To study the geometric properties of the board, as in Section 3.1, it is useful to temporarily ignore the state of the positions and the game dynamics. We thus define the mapping

$$U : (G, E, A, \Delta) \to (G, A).$$

This mapping associates each game with its underlying graph. In this way, properties such as the number of adjacencies $A(x, y)$ can be studied solely from the adjacency structure. Note that different games may determine the same graph, so the mapping is not injective.

3.4 Toward Surrounding Games

Let us see how our theory could take us further than we initially thought.

Othello

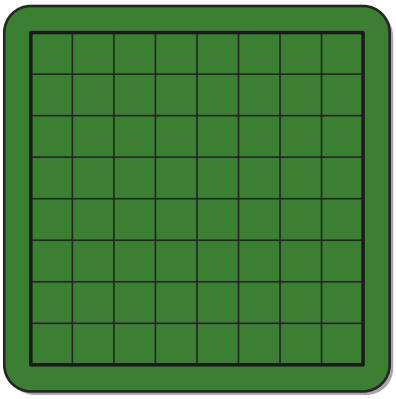


Figure 3.13 – Othello board.

In Figure 3.13, on the Othello board, play takes place inside the squares; this poses no difficulty, as it suffices to think of the intersections as the squares. Unlike Go, stones are adjacent in all directions (diagonals too); we simply need to modify A. Furthermore, in Othello, "capturing" consists of flipping the stones to change their color. In turn, to "surround," it is only necessary to do so in one dimension, that is, to enclose a group of another color along a straight line, either orthogonally or diagonally. We can already accomplish this: it suffices to consider that upon capturing, instead of removing the stones (i.e., assigning $C(i) = 0$), we assign the color of the capturing player (by flipping them). If we had Δ_{Oth}, Othello can be constructed over the game: $(N \times N, \{0, 1, 2\}, A_{ort} \cup A_{diag}, \Delta_{Oth})$. With $|N| = 8$, A_{diag} being the criterion that takes diagonals $(\pm 1, \pm 1)$ into account, and Δ_{Oth} the rules of change in Othello. Thus, the developed construction suggests a possible formalization of *Othello* within the same theoretical framework.

Havannah

Another interesting example is Havannah. Unlike Go, this game takes place on a hexagonal board, and stones are adjacent according to the intrinsic geometry of said board. Once again, it suffices to modify the adjacency criterion. The particularity of Havannah is that there are three distinct ways to win: constructing a ring, a bridge, or a fork (see Figure 3.14).

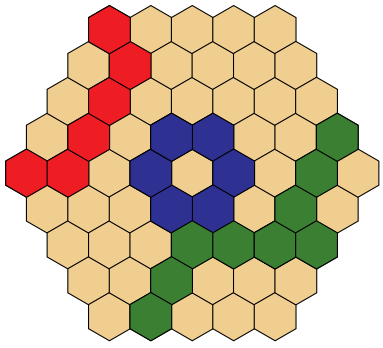


Figure 3.14 – The three winning structures in Havannah: ring, bridge, and fork.

Our theory already accounts for the case of the ring. Indeed, a ring is simply a group that encloses a region. If we distinguish the six corners of the board as special

subsets of positions, a bridge can be described as a group containing stones in at least two distinct corners. Likewise, if we distinguish the six edges of the board, a fork can be described as a group containing stones in at least three distinct edges.

Recalling the properties of $|K(i)|$ discussed in Part I, this suffices for us to identify which points belong to the corners or edges (Section 2.1). In this manner, Havannah's winning condition can be expressed solely in terms of groups, regions, and board subsets. Consequently, the game appears to admit a natural formalization within the same theoretical framework.

Hex

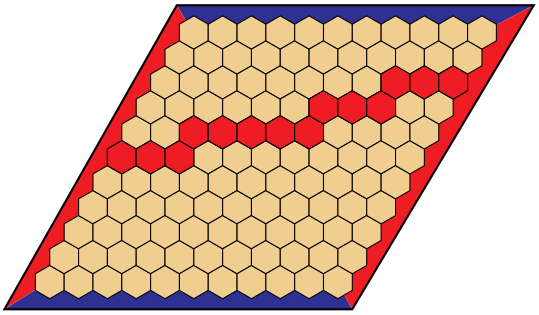


Figure 3.15 – Winning chain connecting two opposite edges.

Let us now consider Hex, Figure 3.15. The board is hexagonal; it suffices to adapt the adjacency criterion. Unlike Go, there are no captures in Hex. To win, one must construct a path from one corner to its opposite. By distinguishing the opposite edges of the board as position subsets, the winning condition can be expressed very simply: a player wins if there exists a group of their color that intersects both opposite edges of their color. Thus, victory in Hex reduces to a property of the groups.

Shannon's Game

Shannon's game constitutes a particularly interesting example; unlike Go, Hex, or Havannah, it does not take place on a board, but rather on an arbitrary finite graph. The two players alternate turns: player Short takes an edge of the graph and colors it, while Cut takes an edge of the graph and removes it. Short wins if their chosen edges contain a path connecting the distinguished vertices A and B. Cut wins if they succeed in disconnecting vertices A and B.

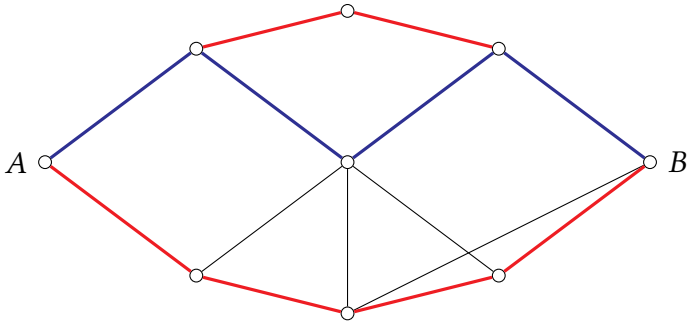


Figure 3.16 – The blue edges belong to Short, the red ones to Cut, and the black ones remain free. The set of blue edges determines a path between the distinguished vertices A and B, so Short wins.

Let $\mathscr{H} = (V_{\mathscr{H}}, E_{\mathscr{H}})$ be a finite graph, and let $A, B \in V_{\mathscr{H}}$ be two distinguished vertices. We consider each edge of $\mathscr{H}$ as a playable position, so the game space is $G_s = E_{\mathscr{H}}$. The state set is $E_s = \{0, 1, 2\}$, where 0 represents a free edge, 1 an edge claimed by Short, and 2 an edge

removed by Cut. Two positions $e_1, e_2 \in G_s$ are adjacent ($e_1 \sim_s e_2$) if and only if they share an endpoint:

$$e_1 \sim_s e_2 \iff e_1 \cap e_2 \neq \emptyset \text{ and } e_1 \neq e_2.$$

Thus, any Shannon switching game on $\mathscr{H}$ is represented by the structure $\Gamma_s = (G_s, E_s, \sim_s, \Delta_s)$.

This example is especially revealing, as it shows that the developed structure does not depend on geometric notions such as regions or territories. Even though it is a game defined on an arbitrary graph, the notions of position, state, adjacency, and group remain sufficient to completely describe its dynamics.

3.5 Conclusion

In this work, we have constructed a theory capable of formally describing classical Go from elementary notions, *where, who, what, and how*, supported by set theory. Throughout this construction, the rules of the game emerge naturally, and intuitive concepts acquire a precise and tractable expression.

However, the principal value of this framework is not limited to formalizing Go. As explored in Part III, the developed structure is sufficiently general to encompass a broader family of games based on the idea of surrounding and capturing. This suggests the existence of a more abstract mathematical object.

Several paths remain open: identifying which properties are exclusive to Go and which belong to the entire family, formalizing this family and seeing how far we can take it, constructing a theory for the infinite Go, or attempting to connect this theory with combinatorial game theory.

Looking back, most of the foundational literature cited in this work stems from the 1960s and 1970s; when surveying existing sources, those pioneering papers contained the precise structural ideas and concepts needed for our model. Many later works approach Go primarily through combinatorial game theory, yet still offer valuable insights into modeling the game itself. Integrating those perspectives to further refine this monograph into a cohesive, accessible, and self-contained treatise remains an exciting direction for future work.

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